



SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (AUTONOMOUS)

(Affiliated to JNTUK, Kakinada), (Recognized by AICTE, New Delhi)

Accredited by NAAC with 'A' Grade

CHINNA AMIRAM (P.O):: BHIMAVARAM :: W.G.Dt., A.P., INDIA :: PIN: 534 204

SCHEME OF INSTRUCTION & EXAMINATION

(Regulation R19)

M.TECH (COMMUNICATION SYSTEMS)

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

(With effect from **2019-2020** Admitted Batch onwards)

I-SEMESTER

Subject Code.	Name of the Subject	Category.	C	L	T	P	Internal Marks	External Marks	Total Marks
M19 CS 1101	Digital Data Communications	PC	3	3	0	0	25	75	100
M19 CS 1102	Advanced Digital Signal Processing	PC	3	3	0	0	25	75	100
#PE-I	Program Elective-I	PE	3	3	0	0	25	75	100
#PE-II	Program Elective-II	PE	3	3	0	0	25	75	100
M19 RD 1101	Research Methodology and IPR	CC	2	2	0	0	25	75	100
M19 CS 1109	Data Communications Lab	PC	2	0	0	4	25	75	100
M19 CS 1110	Advanced Digital Signal Processing Lab	PC	2	0	0	4	25	75	100
#AC-1	Audit course -1	AC	0	2	0	0	0	0	0
Total			18	16	0	8	175	525	700

	Subject Code	Course
#PE-I	M19 CS1103	Radar Signal Processing
	M19 CS 1104	RF Circuit Design
	M19 CS 1105	Advanced Computer Networks
#PE-II	M19 CS 1106	Wireless LANs and PANs
	M19 CS1107	Mobile Computing Technologies
	M19 CS 1108	Network Security & Cryptography
#AC-1 & 2	M19AC0001	English for Research Paper Writing
	M19AC0002	Disaster Management
	M19AC0003	Sanskrit for Technical Knowledge
	M19AC0004	Value Education
	M19AC0005	Constitution of India
	M19AC0006	Pedagogy Studies
	M19AC0007	Stress Management by Yoga
	M19AC0008	Personality Development through Life Enlightenment Skills.

Code	Category	L	T	P	C	I.M	E.M	Exam
M19CS1101	PC	3	0	0	3	25	75	3 Hrs.
DIGITAL DATA COMMUNICATIONS								
Course Objectives: The main objectives of this subject are:								
1	Different modulation techniques to improve the bandwidth and their properties.							
2	Networking and different protocol systems.							
3	Error estimation and correction, asynchronous and synchronous protocols.							
4	Multiplexing techniques, different networking connections and interfacing devices.							
5	Multiple access techniques and analysis.							
Course Outcomes								
S.No	At the end of this course the student will be able to:							Knowledge Level
1	Model digital communication system using appropriate mathematical techniques (error probability, constellation diagrams, phasor diagrams).							K3
2	Understanding the basic concepts of how digital data is transferred across computer networks.							K2
3	Independently understand basic computer network technology.							K2
4	Understand and explain Data Communications System and its components.							K2
5	Identify the different types of network topologies and protocols.							K3
6	Enumerate the layers of the OSI model and TCP/IP. Explain the function(s) of each layer.							K2
7	Identify the different types of network devices and their functions within a network							K3
8	Understand and building the skills of sub netting and routing mechanisms.							K3
9	Familiarity with the basic protocols of computer networks, and how they can be used							K2
10	To assist in network design and implementation.							K6
SYLLABUS								
UNIT-I (10 Hrs)		Digital Modulation Schemes: BPSK, QPSK, 8PSK, 16PSK, 8QAM, 16QAM, DPSK – Methods, Bandwidth Efficiency, Carrier Recovery, Clock Recovery.						
UNIT-II (10 Hrs.)		Basic Concepts of Data Communications, Interfaces and Modems: Data Communication Networks, Protocols and Standards, UART, USB, Line Configuration, Topology, Transmission Modes, Digital Data Transmission, DTE-DCE interface, Categories of Networks – TCP/IP Protocol suite and Comparison with OSI model.						
UNIT-III (8Hrs.)		Error Correction: Types of Errors, Vertical Redundancy Check (VRC), LRC, CRC, Checksum, Error Correction using Hamming code Data Link Control: Line Discipline, Flow Control, Error Control Data Link Protocols:Asynchronous Protocols, Synchronous Protocols, Character Oriented Protocols, Bit-Oriented Protocol, Link Access Procedures.						

UNIT-IV (10 Hrs.)	Multiplexing: Frequency Division Multiplexing (FDM), Time Division Multiplexing (TDM), Multiplexing Application, and DSL. Local Area Networks: Ethernet, Other Ether Networks, Token Bus, Token Ring, FDDI. Metropolitan Area Networks: IEEE 802.6, SMDS Switching: Circuit Switching, Packet Switching, Message Switching. Networking and Interfacing Devices: Repeaters, Bridges, Routers, Gateway, Other Devices.
UNIT-V (10Hrs.)	Multiple Access Techniques: Frequency- Division Multiple Access (FDMA), Time - Division Multiple Access (TDMA), Code - Division Multiple Access (CDMA), OFDM and OFDMA. Random Access, Aloha-Carrier Sense Multiple Access (CSMA)- Carrier Sense Multiple Access with Collision Avoidance (CSMA/CA), Controlled Access- Reservation- Polling- Token Passing, Channelization.
Text Books:	
1.	Data Communication and Computer Networking - B. A.Forouzan, 2 nd Ed., 2003, TMH.
2.	Advanced Electronic Communication Systems - W. Tomasi, 5 th Ed., 2008, PEI.
Reference Book:	
1.	Data Communications and Computer Networks - Prakash C. Gupta, 2006, PHI.
2.	Data and Computer Communications - William Stallings, 8th Ed., 2007, PHI.
3.	Data Communication and Teleprocessing Systems -T. Housely, 2nd Ed, 2008, BSP.
4.	Data Communications and Computer Networks- Brijendra Singh, 2nd Ed., 2005, PHI.

Code	Category	L	T	P	C	I.M	E.M	Exam
M19CS1102	PC	3	0	0	3	25	75	3 Hrs.
ADVANCED DIGITAL SIGNAL PROCESSING								
Course Objectives:								
1	To introduce design and implementation techniques of digital filter (both FIR & IIR).							
2	To introduce the concept of sampling rate conversion, different forms of filter implementation for sampling rate conversion and multistage implementation of sampling rate conversion.							
3	To introduce the applications of sampling rate conversion techniques.							
4	To introduce the concepts of linear prediction and solution of normal equations.							
5	To introduce the Adaptive filter concepts and non-parametric methods of power spectrum estimation.							
6	To introduce wavelets and applications of DSP.							
Course Outcomes:								
S.No	Outcome							Knowledge Level
1	Design and implement the digital filters (both FIR & IIR).							K6
2	Describe sampling rate conversion and multirate signal processing in the digital domain.							K1
3	Apply the concepts of sampling rate conversion in the implementation of digital filter banks, quadrature mirror filters and their use in sub band coding.							K3
4	To understand the theory of forward-backward linear prediction filters and solution of normal equations.							K2
5	Understand Adaptive filtering and the concepts of nonparametric methods of power spectrum estimation.							K2
6	To know the applications based on DSP and multirate DSP.							K3
SYLLABUS								
UNIT-I (10 Hrs)	Review of DFT, FFT, IIR Filters and FIR Filters: Multirate Signal Processing: Introduction, Decimation by a factor D, Interpolation by factor I, Sampling rate conversion by a rational factor I/D, Multistage Implementation of Sampling Rate Conversion, Filter design & Implementation for sampling rate conversion.							
UNIT-II (10 Hrs)	Applications of Multirate Signal Processing: Design of Phase Shifters, Interfacing of Digital Systems with Different Sampling Rates, Implementation of Narrow Band Low Pass Filters, Implementation of Digital Filter Banks, Sub-band Coding of Speech Signals, Quadrature Mirror Filters, Trans-multiplexers, Oversampling A/D and D/A Conversion							
UNIT-III (10 Hrs)	Non-Parametric Methods of Power Spectral Estimation: Estimation of spectra from finite duration observation of signals, Non-parametric Methods: Bartlett, Welch & Blackman-Tukey methods, Comparison of all Non-Parametric methods.							

UNIT-IV (8Hrs)	Implementation of Digital Filters: Introduction to filter structures (IIR & FIR), Frequency sampling structures of FIR, Lattice structures, Forward prediction error, Backward prediction error, Reflection coefficients for lattice realization, Implementation of lattice structures for IIR filters, Advantages of lattice structures.
UNIT-V (10Hrs)	Parametric Methods of Power Spectrum Estimation: Autocorrelation & Its Properties, Relation between autocorrelation & model parameters, AR Models - Yule-Walker & Burg Methods, MA & ARMA models for power spectrum estimation, Finite word length effect in IIR digital Filters – Finite word-length effects in FFT algorithms.
Text Books:	
1.	Digital Signal Processing: Principles, Algorithms & Applications - J.G.Proakis, D. G. Manolakis, 4th Ed., PHI.
2.	Discrete Time Signal Processing - Alan V Oppenheim & R. W Schaffer, PHI.
3.	DSP – A Practical Approach – Emmanuel C. Ifeachor, Barrie. W. Jervis, 2 Ed., Pearson Education.
Reference Books:	
1.	Modern Spectral Estimation: Theory & Application – S. M .Kay, 1988, PHI.
2.	Multi Rate Systems and Filter Banks – P.P.Vaidyanathan – Pearson Education.
3.	Digital Signal Processing – S.Salivahanan, A.Vallavaraj, C.Gnanapriya, 2000,TMH
4.	Digital Spectral Analysis – Jr. Marple

Code	Category	L	T	P	C	I.M	E.M	Exam
M19CS1103	PE	3	0	0	3	25	75	3 Hrs.
RADAR SIGNAL PROCESSING								
(Program Elective-1)								
Course Objectives:								
1	Derivation of Radar range and Design of matched filter for different noises.							
2	Signal detection techniques at receiver.							
3	Optimum Radar Waveforms for Detection of signals in Clutter and various Families.							
4	The characteristics of a Linear pulse and digital compression to Radar signals.							
5	The principles of different phase coding techniques and analysis.							
Course Outcomes								
S.No	Outcome							Knowledge Level
1	Understand the operation of Radar and characteristics of Matched filter for non-white noise.							K2
2	Understand the various detection criterion and types of detectors that can be used to detect the Radar signals in noise.							K2
3	Understand the waveform design requirements and optimum waveforms for the detection of signals in clutter.							K2
4	Know the significance and types of pulse compression techniques for analog and digital signals.							K1
5	Understand the requirements of phase coding in Radar and various Polyphase codes used for phase coding.							K2
SYLLABUS								
UNIT-I (10 Hrs.)	Introduction: Radar Block Diagram, Bistatic Radar, Monostatic Radar, Radar Equation, Information Available from Radar Echo. Review of Radar Range Performance– General Radar Range Equation, Radar Detection with Noise Jamming, Beacon and Repeater Equations, MTI and Pulse Doppler Radar. Matched Filter Receiver – Impulse Response, Frequency Response Characteristic and its Derivation, Matched Filter and Correlation Function, Correlation Detection and Cross-Correlation Receiver, Efficiency of Non-Matched Filters, Matched Filter for Non-White Noise.							
UNIT-II (10 Hrs.)	Detection of Radar Signals in Noise: Detection Criteria – Neyman-Pearson Observer, Likelihood-Ratio Receiver, Inverse Probability Receiver, Sequential Observer, Detectors–Envelope Detector, Logarithmic Detector, I/Q Detector. Automatic Detection-CFAR Receiver, Cell Averaging CFAR Receiver, CFAR Loss, CFAR Uses in Radar. Radar Signal Management–Schematics, Component Parts, Resources and Constraints							

UNIT-III (10 Hrs.)	Waveform Selection: Radar Ambiguity Function and Ambiguity Diagram – Principles and Properties; Specific Cases – Ideal Case, Single Pulse of Sine Wave, Periodic Pulse Train, Single Linear FM Pulse, Noise Like Waveforms, Waveform Design Requirements, Optimum Waveforms for Detection in Clutter, Family of Radar Waveforms.
UNIT-IV (10 Hrs.)	Pulse Compression in Radar Signals: Introduction, Significance, Types, Linear FM Pulse Compression – Block Diagram, Characteristics, Reduction of Time Side lobes, Stretch Techniques, Generation and Decoding of FM Waveforms – Block Schematic and Characteristics of Passive System, Digital Compression, SAW Pulse Compression.
UNIT-V (8 Hrs.)	Phase Coding Techniques: Principles, Binary Phase Coding, Barker Codes, Maximal Length Sequences (MLS/LRS/PN), Block Diagram of a Phase Coded CW Radar. Polyphase Codes : Frank Codes, Costas Codes, Non-Linear FM Pulse Compression, Doppler Tolerant PC Waveforms – Short Pulse, Linear Period Modulation (LPM/HFM), Side lobe Reduction for Phase Coded PC Signals.
Text Books:	
1.	Radar Handbook - M.I. Skolnik, 2nd Ed., 1991, McGraw Hill.
2.	Radar Design Principles: Signal Processing and the Environment - Fred E. Nathanson, 2nd Ed., 1999, PHI.
3.	Introduction to Radar Systems - M.I. Skolnik, 3rd Ed., 2001, TMH.
Reference Books:	
1.	Radar Principles - Peyton Z. Peebles, Jr., 2004, John Wiley.
2.	Radar Signal Processing and Adaptive Systems - R. Nitzberg, 1999, Artech House.

Code	Category	L	T	P	C	I.M	E.M	Exam
M19 CS 1104	PE	3	0	0	3	25	75	3 Hrs.
RF CIRCUIT DESIGN								
(Program Elective-1)								
Course Objectives:								
1	To design and analyse single and multiport networks and RF Filters.							
2	To study the operation and device characteristics of RF Active components.							
3	To design and analyse RF transistor amplifier.							
4	To study the operation of Oscillators and mixers used in RF design							
Course Outcomes								
S.No	Outcome							Knowledge Level
1	Understand the behaviour of RF passive components and model active components.							K2
2	Perform transmission linear analysis.							K4
3	Demonstrate use of Smith Chart for high frequency circuit design.							K2
4	Justify the choice/selection of components from the design aspects.							K5
5	Contribute in the areas of RF circuit design.							K6
SYLLABUS								
UNIT-I (10 Hrs.)	Introduction to RF Electronics: The Electromagnetic Spectrum, units and Physical Constants, Microwave bands – RF behaviour of Passive components: Tuned resonant circuits, Vectors, Inductors and Capacitors - Voltage and Current in capacitor circuits – Tuned RF / IF Transformers.							
UNIT-II (10 Hrs.)	Transmission Line Analysis: Examples of transmission lines- Transmission line equations and Biasing- Micro Strip Transmission Lines- Special Termination Conditions- sourced and Loaded Transmission Lines. Single And Multiport Networks: The Smith Chart, Interconnectivity networks, Network properties and Applications, Scattering Parameters.							
UNIT-III (10 Hrs.)	Matching and Biasing Networks: Impedance matching using discrete components – Micro strip line matching networks, Amplifier classes of Operation and Biasing networks. RF Passive & Active Components: Filter Basics – Lumped filter design – Distributed Filter Design – Diplexer Filters- Crystal and Saw filters- Active Filters - Tunable filters – Power Combiners / Dividers – Directional Couplers – Hybrid Couplers – Isolators. RF Diodes – BJTs- FETs- HEMTs and Models.							
UNIT-IV (10 Hrs.)	RF Transistor Amplifier Design: Characteristics of Amplifiers - Amplifier Circuit Configurations, Amplifier Matching Basics, Distortion and noise products, Stability Considerations, Small Signal amplifier design, Power amplifier design, MMIC amplifiers, Broadband High Power multistage amplifiers, Low noise amplifiers, VGA Amplifiers.							
UNIT-V (8 Hrs.)	Oscillators: Oscillator basics, Low phase noise oscillator design, High frequency Oscillator configuration, LC Oscillators, VCOs, Crystal Oscillators, PLL Synthesizer, and Direct Digital Synthesizer. RF Mixers: Basic characteristics of a mixer - Active mixers- Image Reject and Harmonic mixers, Frequency domain considerations.							

Text Books:	
1.	RF Circuit design: Theory and applications by Reinhold Ludwig, PavelBretchko. Pearson Education Asia Publication, New Delhi 2001.
2.	Radio Frequency and Microwave Communication Circuits – Analysis and Design – Devendra K. Misra, Wiley Student Edition, John Wiley & Sons
Reference Books:	
1.	Radio frequency and Microwave Electronics - Mathew M.Radmangh, 2001, PE Asia Publ.
2.	RF Circuit Design – Christopher Bowick, Cheryl Aljuni and John Biyler, Elsevier Science, 2008.
3.	Secrets of RF Design - Joseph Carr., 3 rd Edition, Tab Electronics.
4.	Complete Wireless Design - Cotter W. Sawyer, 2 nd Edition, Mc-Graw Hill.
5.	Practical RF Circuit Design for Modem Wireless Systems Vol.2 -Less Besser and Rowan Gilmore.

Code	Category	L	T	P	C	I.M	E.M	Exam
M19 CS 1105	PE	3	0	0	3	25	75	3 Hrs.
ADVANCED COMPUTER NETWORKS								
(Program Elective-1)								
Course Objectives:								
1.	The aim of this course is to introduce key concepts and principles of computer networks and Congestion and Quality of Service.							
2.	Illustrate reference models with layers, protocols and interfaces.							
3.	Familiarity with the basic protocols of computer networks, and how they can be used to assist in network design and implementation.							
4.	Have the knowledge on Interconnection Networks, Look into the widely used networks like SONET/SDH and describe how routing protocols work.							
5.	To know the basic concepts of Spread Spectrum, Discuss Ipv6 addresses. Compare Ipv4 and Ipv6 addresses and look into the transition fromIpv4 to Ipv6.							
6.	To have knowledge on IEEE802.11, and to know the basic cellular concepts and describe the features and functions of multiplexing and modulation.							
Course Outcomes								
S.No	Outcome							Knowledge Level
1	Able to Define Congestion and Quality of Service and Illustrate reference models with layers, protocols and interfaces.							K1
2	Familiar with the basic protocols of computer networks, and how they can be used to assist in network design and implementation							K2
3	Understand the general principles behind , addressing, routing, reliable transmission and other stateful protocols							K2
4	Have an informed view of both the internal workings of the Internet and of a number of common Internet applications and protocols							K1
5	Familiar with the basic cellular concepts and understand the importance of multiple accessing schemes.							K2
SYLLABUS								
UNIT-I (10 Hrs)	Congestion and Quality of Service (QoS):Data traffic, Congestion, Congestion Control, Two examples, Quality of Service, Techniques to improve QOS, Integrated Services and Differential services. Queue Management: Passive-Drop trial, Drop front, Random drop, Active- early Random drop, Random Early detection.							
UNIT-II (10 Hrs)	X.25 Standards: X.25 Layers, X.21 Protocol ,Frame Relay: Introduction, Frame relay operation, Frame relay layers, Congestion control, Leaky Bucket algorithms, ATM: Design goals, ATM architecture, Switching, Switch Fabric, ATM layers, Service classes, ATM applications							
UNIT-III (10 Hrs)	Interconnection Networks: Introduction, Banyan Networks, Properties, Crossbar switch, Three Stage Class networks, Rearrange able Networks, Folding algorithm, Benes Networks, Lopping algorithm, Bit allocation algorithm. SONET/SDH: Synchronous Transport signals, Physical configuration, SONET layers, SONET Frame.							

UNIT-IV (8Hrs)	Spread Spectrum: Introduction, Basic concept, Protection against Jamming, Spreading codes(PN sequence), Generation, Properties, Types of Spread Spectrum Modulation, Application of Spread Spectrum. Private Networks: Virtual Private Networks, Network Address Translation Next Generation: IPV6 Transition from IPV4 to IPV6, Mobile IP: Addressing, Agents, Three phases, Inefficiency in Mobile IP.
UNIT-V (10Hrs)	Wireless Networks: Wireless LAN: IEEE802.11, Architecture, MAC Sub Layer, Addressing Mechanism, Physical Layer. Bluetooth: Architecture, Bluetooth layers, Radio layer, Base band layer, L2CAP, Wireless WAN: The Cellular Concept, Cell, Frequency reuse, Principle, Channel Assignment Strategies, Interference and system capacity, Types of interference, Improving capacity in cellular system, Handoff, AMPS, D-AMPS, GSM, CDMA, GPRS, 3G & 4G technologies.
Text Books:	
1.	Data Communication and Networking - B. A. Forouzan, 4 th Ed, TMH
2.	TCP/IP Protocol Suite – B. A. Forouzan, 4 th Ed, TMH
Reference Books:	
1.	Wireless Communication System- Abhishek Yadav, University Sciences Press
2.	Wireless Digital Communications – Kamilofeher, 1999, PHI
3.	High Performance TCP-IP Networking- Mahaboob Hassan, Jain Raj, PHI
4.	ATM Fundamentals- N. N. Biswas, Adventure Book Publishers, 1998
5.	Wireless Communication – T. L. Singhal, McGraw Hill, 2010
6.	Wireless Communication and Networking- Vijay K. Garg, Elsevier, 2009

Code	Category	L	T	P	C	I.M	E.M	Exam
M19 CS 1106	PC	3	0	0	3	25	75	3 Hrs.
WIRELESS LANs AND PANs								
(Program Elective-2)								
Course Objectives:								
1.	To study the different generations of mobile networks.							
2.	To understand the concepts of Wireless LANS.							
3.	To understand the wireless networking IEEE 802.11 for WLAN.							
4.	To understand concepts of different wireless personal area networks							
5.	To study different working group of IEEE 802.15 standards for WLAN.							
Course Outcomes								
S.No	Outcome							Knowledge Level
1	Able to study the different generations of mobile networks.							K1
2	Able to understand the concepts of Wireless LANS.							K2
3	Able to understand the wireless networking IEEE 802.11 for WLAN.							K2
4	Able to understand the concepts of different wireless personal area networks							K2
5	Able to study different working group of IEEE 802.15 standards for WLAN.							K4
SYLLABUS								
UNIT-I (10 Hrs.)	Wireless System & Random Access Protocols: Introduction, First and Second Generation Cellular Systems, Cellular Communications from 1G to 3G, Wireless 4G systems, The Wireless Spectrum; Random Methods: Pure ALOHA, Slotted ALOHA, Carrier Sense Multiple Access (CSMA), Carrier Sense Multiple Access with Collision Detection (CSMA/CD), Carrier Sense Multiple Access with Collision Avoidance (CSMA/CA).							
UNIT-II (10 Hrs.)	Wireless LANs: Introduction, importance of Wireless LANs, WLAN Topologies, Transmission Techniques: Wired Networks, Wireless Networks, comparison of wired and Wireless LANs; WLAN Technologies: Infrared technology, UHF narrowband technology, Spread Spectrum technology.							
UNIT-III (10 Hrs.)	The IEEE 802.11 Standard for Wireless LANs: Network Architecture, Physical layer, The Medium Access Control Layer; MAC Layer issues: Hidden Terminal Problem, Reliability, Collision avoidance, Congestion avoidance, Congestion control, Security, The IEEE 802.11e MAC protocol.							
UNIT-IV (10 Hrs.)	Wireless PANs: Introduction, importance of Wireless PANs, The Bluetooth technology: history and applications, technical overview, the Bluetooth specifications, piconet synchronization and Bluetooth clocks, Master Slave Switch; Bluetooth security; Enhancements to Bluetooth: Bluetooth interference issues, Intra and Inter Pico net scheduling, Bridge selection, Traffic Engineering, QoS and Dynamics Slot Assignment, Scatter net formation.							

UNIT-V (8 Hrs.)	The IEEE 802.15 Working Group for WPANs: The IEEE 802.15.3, The IEEE 802.15.4, ZigBee Technology, ZigBee components and network topologies, The IEEE 802.15.4 LR-WPAN Device architecture: Physical Layer, Data Link Layer, The Network Layer, Applications; IEEE 802.15.3a Ultra wideband
Text Books:	
1.	Ad Hoc and Sensor Networks - Carlos de MoraisCordeiro and Dharma Prakash Agrawal, World Scientific, 2011.
2.	Wireless Communications and Networking - Vijay K.Garg, Morgan Kaufmann Publishers, 2009.
Reference Books:	
1.	Wireless Networks - KavehPahlaram, Prashant Krishnamurthy, PHI, 2002.
2.	Wireless Communication- Marks Ciampor, JeorgeOlenewa, Cengage Learning, 2007.

Code	Category	L	T	P	C	I.M	E.M	Exam
M19 CS 1107	PE	3	0	0	3	25	75	3 Hrs.
MOBILE COMPUTING TECHNOLOGIES								
(Program Elective-2)								
Course Objectives:								
1.	To acquire Knowledge of WAP, Wireless Lan							
2.	To study client programming and different OS architectures.							
3.	To learn J2ME technology and Mobile VoIP							
4.	To become familiar with Security issues in Mobile Computing							
Course Outcomes								
S.No	Outcome							Knowledge Level
1	Apply advanced data communication methods and networking protocols for wireless and mobile environments							K3
2	Utilize and employ application frameworks for developing mobile applications including under disconnected and weakly connected environment							K3
3	Create web sites suitable for mobile environments							K6
4	Select components and networks for particular application							K3
5	Creatively analyse mobile and wireless networks							K4
6	Critically analyse security issues of mobile and wireless computing systems							K4
SYLLABUS								
UNIT-I (10 Hrs)	Introduction to Mobile Computing Architecture: Mobile Computing – Dialog Control – Networks – Middleware and Gateways – Application and Services – Developing Mobile Computing Applications – Security in Mobile Computing – Architecture for Mobile Computing – Three Tier Architecture – Design considerations for Mobile Computing – Mobile Computing through Internet – Making existing Applications Mobile Enabled.							
UNIT-II (10 Hrs)	Cellular Technologies: GSM, GPS, GPRS, CDMA and 3G: Bluetooth – Radio Frequency Identification – Wireless Broadband – Mobile IP – Internet Protocol Version 6 (IPv6) – Java Card – GSM Architecture – GSM Entities – Call Routing in GSM – PLMN Interfaces – GSM addresses and Identifiers – Network aspects in GSM – Authentication and Security – Mobile computing over SMS – GPRS and Packet Data Network – GPRS Network Architecture – GPRS Network Operations – Data Services in GPRS – Applications for GPRS – Limitations of GPRS – Spread Spectrum technology – Is-95 – CDMA Versus GSM – Wireless Data – Third Generation Networks – Applications on 3G							
UNIT-III (10 Hrs)	Wireless Application Protocol (WAP) and Wireless LAN: WAP – MMS – Wireless LAN Advantages – IEEE 802.11 Standards – Wireless LAN Architecture – Mobility in wireless LAN Intelligent Networks and Interworking: Introduction – Fundamentals of Call processing – Intelligence in the Networks – SS#7 Signalling – IN Conceptual Model (INCM) – soft switch – Programmable Networks – Technologies and Interfaces for IN							

UNIT-IV (8Hrs)	Client Programming, Palm OS, Symbian OS, Win CE Architecture: Introduction – Moving beyond the Desktop – A Peek under the Hood: Hardware Overview – Mobile phones – PDA – Design Constraints in Applications for Handheld Devices – Palm OS architecture – Application Development – Multimedia – Symbian OS Architecture – Applications for Symbian, Different flavors of Windows CE -Windows CE Architecture. J2ME: JAVA in the Handset – The Three-prong approach to JAVA Everywhere – JAVA 2 Micro Edition (J2ME) technology – Programming for CLDC – GUI in MIDP – UI Design Issues – Multimedia – Record Management System – Communication in MIDP – Security considerations in MIDP – Optional Packages
UNIT-V (10Hrs)	Voice Over Internet Protocol and Convergence: Voice over IP- H.323 Framework for Voice over IP – Session Initiation Protocol – Comparison between H.323 and SIP – Real Time protocols – Convergence Technologies – Call Routing – Voice over IP Applications – IP multimedia subsystem (IMS) – Mobile VoIP Security Issues in Mobile Computing: Introduction – Information Security – Security Techniques and Algorithms – Security Protocols– Public Key Infrastructure – Trust – Security Models – Security frameworks for Mobile Environment
Text Books:	
1.	Mobile Computing – Technology, Applications and Service Creation – Asoke K Talukder, Roopa R Yavagal, 2009, TATA McGraw Hill
2.	Mobile Communications – Jochen Schiller – 2 nd Edition – Pearson Education
Reference Books:	
1.	The CDMA 2000 System for Mobile Communications – VieriVaughni, Alexander Damn Jaonvic – Pearson
2.	Adalestein - Fundamentals of Mobile & Pervasive Computing, 2008, TMH

Code	Category	L	T	P	C	I.M	E.M	Exam
M19 CS 1108	PE	3	0	0	3	25	75	3 Hrs.
NETWORK SECURITY & CRYPTOGRAPHY								
(Program Elective-2)								
Course Objectives:								
1.	To give a clear insight into cryptography, authentication and emerging security standards.							
2.	To impart knowledge on network security protocols.							
Course Outcomes								
S.No	Outcome							Knowledge Level
1	Identify and utilize different forms of cryptography techniques.							K3
2	Incorporate authentication and security in the network applications.							K6
3	Distinguish among different types of threats to the system and handle the same.							K4
4	To become familiar with Security issues in Electronic transaction.							K2
SYLLABUS								
UNIT-I (10 Hrs)	Introduction: Attacks, Services and Mechanisms, Security attacks, Security services, A Model for Internetwork security. Classical Techniques: Conventional Encryption model, Steganography, Classical Encryption Techniques. Modern Techniques: Simplified DES, Block Cipher Principles, Data Encryption standard, Strength of DES, Differential and Linear Cryptanalysis, Block Cipher Design Principles and Modes of operations.							
UNIT-II (10 Hrs)	Encryption Algorithms: Triple DES, International Data Encryption algorithm, Blowfish, RC5, CAST-128, RC2, Characteristics of Advanced Symmetric block ciphers. Conventional Encryption :Placement of Encryption function, Traffic confidentiality, Key distribution, Random Number Generation.							
UNIT-III (10 Hrs)	Public Key Cryptography: Principles, RSA Algorithm, Key Management, Diffie-Hellman Key exchange, Elliptic Curve Cryptography. Number Theory: Prime and Relatively prime numbers, Modular arithmetic, Fermat’s and Euler’s theorems, Testing for primality, Euclid’s Algorithm, the Chinese remainder theorem, Discrete logarithms.							
UNIT-IV (10 Hrs)	Message Authentication and Hash Functions: Authentication requirements and functions, Message Authentication, Hash functions, Security of Hash functions and MACs. Hash and Mac Algorithms: MD File, Message digest Algorithm, Secure Hash Algorithm, RIPEMD-160, HMAC. Digital signatures and Authentication protocols: Digital signatures, Authentication Protocols, Digital signature standards. Authentication Applications: Kerberos, X.509 directory Authentication service. Electronic Mail Security: Pretty Good Privacy, S/MIME.							

UNIT-V (8 Hrs)	IP Security: Overview, Architecture, Authentication, Encapsulating Security Payload, Combining security Associations, Key Management. Web Security: Web Security requirements, Secure sockets layer and Transport layer security, Secure Electronic Transaction. Intruders, Viruses and Worms: Intruders, Viruses and Related threats. Fire Walls: Firewall Design Principles, Trusted systems.
Text Books:	
1.	Cryptography and Network Security: Principles and Practice - William Stallings, Pearson Education.
2.	Network Security Essentials (Applications and Standards) by William Stallings Pearson Education.
Reference Books:	
1.	Fundamentals of Network Security by Eric Maiwald (Dreamtech press)
2.	Network Security - Private Communication in a Public World by Charlie Kaufman, Radia Perlman and Mike Speciner, Pearson/PHI.
3.	Principles of Information Security, Whitman, Thomson.
4.	Network Security: The complete reference, Robert Bragg, Mark Rhodes, TMH
5.	Introduction to Cryptography, Buchmann, Springer.

Code	Category	L	T	P	C	I.M	E.M	Exam
M19RD1101	RD	2	0	0	2	25	75	3 Hrs.
RESEARCH METHODOLOGY AND IPR								
(Common to CST,CS,PSA, IT & CAD)								
Course Objectives:								
1.	To bring awareness on Research Methodology and research ethics.							
2.	Familiarize the concepts of IPR.							
Course Outcomes								
S.No	Outcome							Knowledge Level
1.	Analyze research related information							K4
2.	Formulate a Research Proposals and Publish papers with research ethics							K6
3.	Award for Intellectual Property Rights like Patents, Trade and Copyrights							K5
4.	Analyze Various Intellectual Property Rights							K4
5.	AssessNew Developments of IPRs in National and International level							K5
SYLLABUS								
UNIT-I (6 Hrs)	Meaning of research problem, Sources of research problem, Criteria Characteristics of a good research problem, Errors in selecting a research problem, Scope and objectives of research problem. Approaches of investigation of solutions for research problem, data collection, analysis, interpretation, Necessary instrumentations							
UNIT-II (6 Hrs)	Effective literature studies approaches, analysis Plagiarism, Research ethics, Effective technical writing, how to write report, Paper Developing a Research Proposal, Format of research proposal, a presentation and assessment by a review committee							
UNIT-III (6 Hrs)	Nature of Intellectual Property: Patents, Designs, Trade and Copyright. Process of Patenting and Development: technological research, innovation, patenting, development. International Scenario: International cooperation on Intellectual Property. Procedure for grants of patents, Patenting under PCT.							
UNIT-IV (4 Hrs)	Patent Rights: Scope of Patent Rights. Licensing and transfer of technology. Patent information and databases. Geographical Indications.							
UNIT-V (6 Hrs)	New Developments in IPR: Administration of Patent System. New developments in IPR; IPR of Biological Systems, Computer Software etc. Traditional knowledge Case Studies, IPR and IITs.							
Text Books:								
1.	Stuart Melville and Wayne Goddard, —Research methodology: an introduction for science & engineering students’l							
2.	Wayne Goddard and Stuart Melville, —Research Methodology: An Introductionll							
3.	Ranjit Kumar, 2nd Edition, —Research Methodology: A Step by Step Guide for beginnersll							
Reference Books:								
1.	Halbert, —Resisting Intellectual Propertyll, Taylor & Francis Ltd, 2007.							
2.	Mayall, —Industrial Designll, McGraw Hill, 1992.							
3.	Niebel, —Product Designll, McGraw Hill, 1974.							

4.	Asimov, —Introduction to Design, Prentice Hall, 1962.
5.	Robert P. Merges, Peter S. Menell, Mark A. Lemley, — Intellectual Property in New Technological Age, 2016.
6.	T. Ramappa, —Intellectual Property Rights Under WTO, S. Chand, 2008

Code	Category	L	T	P	C	I.M	E.M	Exam
M19 CS 1109	PC	0	0	4	2	25	75	3 Hrs.
DATA COMMUNICATIONS LAB								
Course Objectives:								
1.	To study the analog& digital link set up using a fiber in detail							
2.	Understand the pc-to-pc communication using parallel port							
3.	To study of LAN using star topology, bus topology and tree topology and configure modem of a computer							
Course Outcomes								
S.No	Outcome							Knowledge Level
1.	Students can identify the type of fibre optical cable and test their applications							K3
2.	Understand the fundamental concepts of data communications and networking							K2
3.	Identify different components and their respective roles in a computer communication system.							K3
4.	Acquaint them-selves with networking simulation tools, configuring of networking devices and understand their functionality.							K2
List of Experiments								
1.	Study of serial interface RS – 232							
2.	Study of pc to pc communication using parallel port							
3.	To establish pc-pc communication using LAN							
4.	Study of LAN using star topology, bus topology and tree topology							
5.	Study and configure modem of a computer							
6.	To configure a hub/switch							
7.	To study the interconnections of cables for data communication							
8.	Study of a wireless communication system							
9.	Set up of time division multiplexing using fiber optics							
10.	Digital Fibre Optical Transmitter and Receiver							
Reference Books:								
1.	Optical Fibre Communications – Gerd Keiser, 3 rd Ed., 2000, McGraw Hill.							
2.	Optical Fibre Communications: Principles and Practice – John.M.Senior, 2 nd Ed., 2000, PE.							
3.	Data Communications and Computer Networks - Prakash C. Gupta, 2006, PHI.							
4.	Data and Computer Communications - William Stallings, 8 th Ed., 2007, PHI.							

Code	Category	L	T	P	C	I.M	E.M	Exam
M19 CS 1110	PC	0	0	4	2	25	75	3 Hrs.
ADVANCED DIGITAL SIGNAL PROCESSING LAB								
Course Objectives:								
1.	To design different filters in software							
2.	To apply transforms in time and frequency							
3.	To perform decimation and interpolation							
Course Outcomes								
S.No	Outcome							Knowledge Level
1	Design different digital filters in software							K6
2	Apply various transforms in time and frequency							K3
3	Perform decimation and interpolation							K6
LIST OF EXPERIMENTS								
1.	Basic Signal Representation							
2.	Correlation Auto And Cross							
3.	Stability Using Hurwitz Routh Criteria							
4.	Sampling FFT Of Input Sequence							
5.	Butterworth Low pass And High pass Filter Design							
6.	Chebychev Type I,II Filter							
7.	State Space Matrix from Differential Equation							
8.	Normal Equation Using Levinson Durbin							
9.	Decimation And Interpolation Using Rational Factors							
10.	Maximally Decimated Analysis DFT Filter							
11.	Cascade Digital IIR Filter Realization							
12.	Convolution And M Fold Decimation & PSD Estimator							
13.	Estimation Of PSD							
14.	Inverse Z Transform							
15.	Group Delay Calculation							
16.	Separation of T/F							
17.	Parallel Realization of IIR filter							
Text Books:								
1.	Discrete Time Signal Processing - Alan V Oppenheim & R. W Schaffer, PHI.							
2.	DSP – A Practical Approach – Emmanuel C. Ifeachor, Barrie. W. Jervis, 2 Ed., Pearson Education							
Reference Books:								
1.	Modern Spectral Estimation: Theory & Application – S. M .Kay, 1988, PHI.							
2.	Multi Rate Systems and Filter Banks – P.P.Vaidyanathan – Pearson Education.							
3.	Digital Signal Processing – S.Salivahanan, A.Vallavaraj, C.Gnanapriya, 2000,TMH							
4.	Digital Spectral Analysis – Jr. Marple							

Code	Category	L	T	P	C	I.M	E.M	Exam
M19AC0001	AC	2	0	0	0	0	0	--
ENGLISH FOR RESEARCH PAPER WRITING								
Course Objectives:								
1.	Understand how to improve your writing skills and level of readability							
2.	Learn about what to write in each section.							
3.	Understand the skills needed when writing a Title.							
4.	Ensure the good quality of paper at very first-time submission							
Course Outcomes								
S.No	Outcome							Knowledge Level
1.	Understand that how to improve your writing skills and level of readability							K2
2.	Learn about what to write in each section							K2
3.	Understand the skills needed when writing a Title Ensure the good quality of paper at very first time submission							K2
SYLLABUS								
UNIT-I (4Hrs)	Planning and Preparation, Word Order, Breaking up long sentences, Structuring Paragraphs and Sentences, Being Concise and Removing Redundancy, Avoiding Ambiguity and Vagueness							
UNIT-II (4Hrs)	Clarifying Who Did What, Highlighting Your Findings, Hedging And Criticizing, Paraphrasing and Plagiarism, Sections of a Paper.							
UNIT-III (4Hrs)	Abstracts, Introduction, Review of the Literature, Methods, Results, Discussion, Conclusions, The Final Check.							
UNIT-IV (4Hrs)	Key skills are needed when writing a Title, key skills are needed when writing an abstract, key skills are needed when writing an introduction, skills needed when writing a Review of the Literature, skills are needed when writing the Methods.							
UNIT-V (4Hrs)	skills needed when writing the Results, skills are needed when writing the Discussion, skills are needed when writing the Conclusions, useful phrases, how to ensure paper is as good as it could possibly be the first- time submission							
Text Books:								
1.	Goldbort R (2006) Writing for Science, Yale University Press (available on Google Books)							
2.	Day R (2006) How to Write and Publish a Scientific Paper, Cambridge University Press							
Reference Books:								
1.	Highman N (1998), Handbook of Writing for the Mathematical Sciences, SIAM. Highman’s book.							
2.	Adrian Wallwork , English for Writing Research Papers, Springer New York Dordrecht Heidelberg London, 2011							

Code	Category	L	T	P	C	I.M	E.M	Exam
M19AC0002	AC	2	0	0	0	0	0	--
DISASTER MANAGEMENT								
Course Objectives:								
1.	Learn to demonstrate a critical understanding of key concepts in disaster risk reduction and humanitarian response.							
2.	Critically evaluate disaster risk reduction and humanitarian response policy and practice from multiple perspectives.							
3.	Develop an understanding of standards of humanitarian response and practical relevance in specific types of disasters and conflict situations.							
4.	Critically understand the strengths and weaknesses of disaster management approaches, planning & programming in different countries, particularly their home country or the countries they work in.							
SYLLABUS								
UNIT-I (4Hrs)	Disaster: Definition, Factors And Significance; Difference Between Hazard And Disaster; Natural And Manmade Disasters: Difference, Nature, Types And Magnitude.							
UNIT-II (4Hrs)	Economic Damage, Loss Of Human And Animal Life, Destruction Of Ecosystem. Natural Disasters: Earthquakes, Volcanisms, Cyclones, Tsunamis, Floods, Droughts And Famines, Landslides And Avalanches, Man-made disaster: Nuclear Reactor Meltdown, Industrial Accidents, Oil Slicks And Spills, Outbreaks Of Disease And Epidemics, War And Conflicts							
UNIT-III (4Hrs)	Disaster Prone Areas In India: Study Of Seismic Zones; Areas Prone To Floods And Droughts, Landslides And Avalanches; Areas Prone To Cyclonic And Coastal Hazards With Special Reference To Tsunami; Post-Disaster Diseases And Epidemics.							
UNIT-IV (4Hrs)	Disaster Preparedness And Management Preparedness: Monitoring Of Phenomena Triggering A Disaster Or Hazard; Evaluation Of Risk: Application Of Remote Sensing, Data From Meteorological And Other Agencies, Media Reports: Governmental And Community Preparedness.							
UNIT-V (4Hrs)	Risk Assessment Disaster Risk: Concept And Elements, Disaster Risk Reduction, Global And National Disaster Risk Situation. Techniques Of Risk Assessment, Global Co-Operation In Risk Assessment And Warning, People's Participation In Risk Assessment. Strategies for Survival. Disaster Mitigation Meaning, Concept And Strategies Of Disaster Mitigation, Emerging Trends In Mitigation. Structural Mitigation And Non-Structural Mitigation, Programs Of Disaster Mitigation In India.							
Text Books:								
1.	R. Nishith, Singh AK, Disaster Management in India: Perspectives, issues and strategies New Royal Book Company.							
2.	Sahni, PardeepEt.Al. (Eds.),Disaster Mitigation Experiences And Reflections, Prentice Hall Of India, New Delhi.							
3.	Goel S. L. Disaster Administration And Management Text And Case Studies ,Deep &Deep Publication Pvt. Ltd., New Delhi.							

Code	Category	L	T	P	C	I.M	E.M	Exam
M19AC0003	AC	2	0	0	0	0	0	--
SANSKRIT FOR TECHNICAL KNOWLEDGE								
Course Objectives:								
1.	To get a working knowledge in illustrious Sanskrit, the scientific language in the world							
2.	Learning of Sanskrit to improve brain functioning							
3.	Learning of Sanskrit to develop the logic in mathematics, science & other subjects enhancing the memory power							
4.	The engineering scholars equipped with Sanskrit will be able to explore the huge knowledge from ancient literature							
Course Outcomes								
S.No	Outcome							Knowledge Level
1	Understanding basic Sanskrit language.							K2
2	Ancient Sanskrit literature about science & technology can be understood.							K2
3	Being a logical language will help to develop logic in students.							K6
SYLLABUS								
UNIT-I (7Hrs)	Alphabets in Sanskrit, Past/Present/Future Tense, Simple Sentences							
UNIT-II (7Hrs)	Order, Introduction of roots, Technical information about Sanskrit Literature							
UNIT-III (7Hrs)	Technical concepts of Engineering-Electrical, Mechanical, Architecture, Mathematics							
Text Books:								
1.	“AbhyasaPustakam” – Dr.Vishwas, Samskrita-Bharati Publication, New Delhi							
2.	“Teach Yourself Sanskrit” PrathamaDiksha-VempatiKutumbasastry, Rashtriya Sanskrit Sansthanam, New Delhi Publication							
3.	“India’s Glorious Scientific Tradition” Suresh Soni, Ocean books (P) Ltd., New Delhi							

Code	Category	L	T	P	C	I.M	E.M	Exam
M19AC0004	AC	2	0	0	0	0	0	--
VALUE EDUCATION								
Course Objectives:								
1.	Understand the value of education and self- development							
2.	Imbibe good values in students							
3.	Let should know about the importance of character							
Course Outcomes								
S.No	Outcome							Knowledge Level
1	Knowledge of self-development							K1
2	Learn the importance of Human values							K2
3	Developing the overall personality							K3
SYLLABUS								
UNIT-I (4Hrs)	Values and self-development –Social values and individual attitudes. Work ethics, Indian vision of humanism.Moral and non- moral valuation. Standards and principles. Value judgements.							
UNIT-II (4Hrs)	Importance of cultivation of values. Sense of duty. Devotion, Self-reliance. Confidence, Concentration. Truthfulness, Cleanliness. Honesty, Humanity. Power of faith, National Unity, Patriotism, Love for nature ,Discipline							
UNIT-III (4Hrs)	Personality and Behavior Development - Soul and Scientific attitude. Positive Thinking. Integrity and discipline. Punctuality, Love and Kindness. Avoid fault Thinking. Free from anger, Dignity of labour. Universal brotherhood and religious tolerance. True friendship. Happiness Vs suffering, love for truth. Aware of self-destructive habits. Association and Cooperation, Doing best for saving nature							
UNIT-IV (4Hrs)	Character and Competence –Holy books vs Blind faith.Self-management and Good health. Science of reincarnation. Equality, Nonviolence ,Humility, Role of Women. All religions and same message. Mind your Mind, Self-control. Honesty, Studying effectively.							
Text Books:								
1.	Chakroborty, S.K. “Values and Ethics for organizations Theory and practice”, Oxford University Press, New Delhi							

Code	Category	L	T	P	C	I.M	E.M	Exam
M19AC0005	AC	2	0	0	0	0	0	--
CONSTITUTION OF INDIA								
Course Objectives:								
1.	Understand the premises informing the twin themes of liberty and freedom from a civil rights perspective.							
2.	To address the growth of Indian opinion regarding modern Indian intellectuals’ constitutional role and entitlement to civil and economic rights as well as the emergence of nationhood in the early years of Indian nationalism							
3.	To address the role of socialism in India after the commencement of the Bolshevik Revolution in 1917 and its impact on the initial drafting of the Indian Constitution							
Course Outcomes								
S.No	Outcome							Knowledge Level
1	Discuss the growth of the demand for civil rights in India for the bulk of Indians before the arrival of Gandhi in Indian politics.							K2
2	Discuss the intellectual origins of the framework of argument that informed the conceptualization of social reforms leading to revolution in India.							K2
3	Discuss the circumstances surrounding the foundation of the Congress Socialist Party [CSP] under the leadership of Jawaharlal Nehru and the eventual failure of the proposal of direct elections through adult suffrage in the Indian Constitution.							K2
4	Discuss the passage of the Hindu Code Bill of 1956.							K2
SYLLABUS								
UNIT-I (4Hrs)	History of Making of the Indian Constitution: History , Drafting Committee, (Composition & Working)							
UNIT-II (4Hrs)	Philosophy of the Indian Constitution: Preamble ,Salient Features							
UNIT-III (4Hrs)	Fundamental Rights,Right to Equality, Right to Freedom,Right against Exploitation, Right to Freedom of Religion,Cultural and Educational Rights,Right to Constitutional Remedies, Directive Principles of State Policy, Fundamental Duties							
UNIT-IV (4Hrs)	Organs of Governance: Parliament, Composition, Qualifications and Disqualifications ,Powers and Functions, Executive, President , Governor, Council of Ministers, Judiciary, Appointment and Transfer of Judges, Qualifications, Powers and Functions Local Administration: District’s Administration head: Role and Importance, Municipalities: Introduction, Mayor and role of Elected Representative, CEO of Municipal Corporation, Pachayati raj: Introduction, PRI: Zilla Panchayat. Elected officials and their roles, CEO Zilla Panchayat: Position and role. Block level: Organizational Hierarchy (Different departments),Village level: Role of Elected and Appointed officials, Importance of grass root democracy							

UNIT-V (4Hrs)	Election Commission: Election Commission: Role and Functioning. Chief Election Commissioner and Election Commissioners. State Election Commission: Role and Functioning. Institute and Bodies for the welfare of SC/ST/OBC and women.
Text Books:	
1.	The Constitution of India, 1950 (Bare Act), Government Publication.
2.	Dr. S. N. Busi, Dr. B. R. Ambedkar framing of Indian Constitution, 1st Edition, 2015.
3.	M. P. Jain, Indian Constitution Law, 7th Edn., LexisNexis, 2014.
4.	D.D. Basu, Introduction to the Constitution of India, LexisNexis, 2015.

Code	Category	L	T	P	C	I.M	E.M	Exam
M19AC0006	AC	2	0	0	0	0	0	--
PEDAGOGY STUDIES								
SYLLABUS								
UNIT-I (4Hrs)	Introduction and Methodology: Aims and rationale, Policy background, Conceptual framework and terminology Theories of learning, Curriculum, Teacher education. Conceptual framework, Research questions. Overview of methodology and Searching.							
UNIT-II (4Hrs)	Thematic overview: Pedagogical practices are being used by teachers in formal and informal classrooms in developing countries, Curriculum, Teacher education.							
UNIT-III (4Hrs)	Evidence on the effectiveness of pedagogical practices, Methodology for the in depth stage: quality assessment of included studies. How can teacher education (curriculum and practicum) and the school curriculum and guidance materials best support effective pedagogy.							
UNIT-IV (4Hrs)	Theory of change, Strength and nature of the body of evidence for effective pedagogical practices Pedagogic theory and pedagogical approaches. Teachers’ attitudes and beliefs and Pedagogic strategies. Professional development: alignment with classroom practices and follow-up support							
UNIT-V (4Hrs)	Research gaps and future directions: Research design, Contexts, Pedagogy, Teacher education, Curriculum and assessment, Dissemination and research impact.							
Text Books:								
1.	Ackers J, Hardman F (2001) Classroom interaction in Kenyan primary schools, Compare, 31 (2): 245-261.							
2.	Agrawal M (2004) Curricular reform in schools: The importance of evaluation, Journal of Curriculum Studies, 36 (3): 361-379.							
3.	Akyeampong K (2003) Teacher training in Ghana - does it count? Multi-site teacher education research project (MUSTER) country report 1. London: DFID.							
4.	Akyeampong K, Lussier K, Pryor J, Westbrook J (2013) Improving teaching and learning of basic maths and reading in Africa: Does teacher preparation count? International Journal of Educational Development, 33 (3): 272–282.							
5.	Alexander RJ (2001) Culture and pedagogy: International comparisons in primary education. Oxford and Boston: Blackwell.							
6.	Chavan M (2003) Read India: A mass scale, rapid, „learning to read“ campaign.							
7.	www.pratham.org/images/resource%20working%20paper%202.pdf.							

Code	Category	L	T	P	C	I.M	E.M	Exam
M19AC0007	AC	2	0	0	0	0	0	--
STRESS MANAGEMENT BY YOGA								
Course Objectives:								
1.	To achieve overall health of body and mind							
2.	To overcome stress							
Course Outcomes								
S.No	Outcome							Knowledge Level
1	Develop a healthy mind in a healthy body thus improving social health also.							K2
2	Improve efficiency							K2
SYLLABUS								
UNIT-I (7Hrs)	Definitions of Eight parts of yoga (Ashtanga)							
UNIT-II (7Hrs)	Yam and Niyam. Do's and Don'ts in life. i) Ahinsa, satya, astheya, brahmacharya and aparigraha ii) Shaucha, santosh, tapa, swadhyay, ishvarapranidhana							
UNIT-III (7Hrs)	Asan and Pranayam i) Various yog poses and their benefits for mind & body ii)Regularization of breathing techniques and its effects-Types pranayama							
Text Books:								
1.	‘Yogic Asanas for Group Training-Part-I’ : Janardan Swami YogabhyasiMandal, Nagpur							
2.	“Raja Yoga or conquering the Internal Nature” by Swami Vivekananda, AdvaitaAshrama (Publication Department), Kolkata							

Code	Category	L	T	P	C	I.M	E.M	Exam
M19AC0008	AC	2	0	0	0	0	0	--
PERSONALITY DEVELOPMENT THROUGH LIFEENLIGHTENMENT SKILLS								
Course Objectives:								
1.	To learn to achieve the highest goal happily							
2.	To become a person with a stable mind, pleasing personality and determination							
3.	To awaken wisdom in students							
Course Outcomes								
S.No	Outcome							Knowledge Level
1	Study of Shrimad-Bhagwad-Geeta will help the student in developing his personality and achieve the highest goal in life.							K2
2	The person who has studied Geeta will lead the nation and mankind to peace and prosperity.							K2
3	Study of Neetishatakam will help in developing versatile personality of students.							K2
SYLLABUS								
UNIT-I (7Hrs)	Neetisatakam-Holistic development of personality Verses- 19,20,21,22 (wisdom), Verses- 29,31,32 (pride & heroism) Verses- 26,28,63,65 (virtue),Verses- 52,53,59 (don'ts),Verses- 71,73,75,78 (do's)							
UNIT-II (7Hrs)	Approach to day to day work and duties. ShrimadBhagwadGeeta : Chapter 2-Verses: 41, 47,48 Chapter 3-Verses 13, 21, 27, 35, Chapter 6-Verses 5,13,17, 23, 35, Chapter 18-Verses 45, 46, 48.							
UNIT-III (7Hrs)	Statements of basic knowledge, ShrimadBhagwadGeeta: Chapter2-Verses 56, 62, 68, Chapter 12 -Verses 13, 14, 15, 16,17, 18, Personality of Role model. ShrimadBhagwadGeeta: Chapter2-Verses 17, Chapter 3-Verses 36,37,42, Chapter 4-Verses 18, 38,39, Chapter18 – Verses 37,38,63							
Text Books:								
1.	‘Srimad Bhagavad Gita” by Swami SwarupanandaAdvaita Ashram (Publication Department), Kolkata							
2.	Bhartrihari’s Three Satakam (Niti-sringar-vairagya) by P.Gopinath,Rashtriya Sanskrit Sansthanam, New Delhi.							



Estd:1980

**SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE
(AUTONOMOUS)**

(Affiliated to JNTUK, Kakinada), (Recognized by AICTE, New Delhi)

Accredited by NAAC with 'A' Grade

CHINNA AMIRAM (P.O):: BHIMAVARAM :: W.G.Dt., A.P., INDIA :: PIN: 534 204

**SCHEME OF INSTRUCTION & EXAMINATION
(Regulation R19)**

M.TECH –COMMUNICATION SYSTEMS

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

(With effect from 2019-2020 Admitted Batch onwards)

II-SEMESTER

Subject code	Name of the Subject	Category	C	L	T	P	Internal Marks	External Marks	Total Marks
M19 CS 1201	Wireless Communications and Networks	PC	3	3	0	0	25	75	100
M19 CS 1202	Image and Video Processing	PC	3	3	0	0	25	75	100
#PE-III	Elective-III	PE	3	3	0	0	25	75	100
#PE-IV	Elective-IV	PE	3	3	0	0	25	75	100
M19 CS 1209	Advanced Communications Lab	PC	2	0	0	4	25	75	100
M19 CS 1210	Advanced Digital Image and Video Processing Laboratory	PC	2	0	0	4	25	75	100
M19 CS 1211	Mini Project with Seminar	PC	2	0	0	4	100	0	100
#AC-2	Audit Course 2	AC	0	2	0	0	0	0	0
Total			18	14	0	12	250	450	700

	Course Code	Course
#PE-III	M19 CS 1203	Soft Computing Techniques
	M19 CS 1204	Internet Protocols
	M19 CS 1205	Cyber Security
#PE-IV	M19 CS 1206	Optical Networks
	M19 CS 1207	Digital Signal Processors and Architectures
	M19 CS 1208	Radio Navigational Aids

Code	Category	L	T	P	C	I.M	E.M	Exam
M19CS1201	PC	3	0	0	3	25	75	3 Hrs.
WIRELESS COMMUNICATIONS AND NETWORKS								
Course Objectives: The main objectives of this subject are:								
1	An understanding on functioning of wireless communication system and evolution of different wireless communication systems and standards.							
2	An ability to compare recent technologies used for wireless communication.							
3	An ability to explain the architecture, functioning, protocols, capabilities and application of various wireless communication networks							
4	An ability to explain multiple access techniques for Wireless Communication							
5	An ability to evaluate design challenges, constraints and security issues associated with Ad-hoc wireless networks.							
Course Outcomes								
S.No	At the end of this course the student will be able to:							Knowledge Level
1	Understand Cellular communication concepts							K1
2	Study the mobile radio propagation							K1
3	Study the wireless networks and different types of MAC protocols							K1
SYLLABUS								
UNIT-I (10 Hrs)	The Cellular Concept-System Design Fundamentals: Introduction, Frequency Reuse, Interference and system capacity – Co channel Interference and system capacity, Channel planning for Wireless Systems, Adjacent Channel interference , Power Control for Reducing interference, Improving Coverage & Capacity in Cellular Systems- Cell Splitting, Sectoring, Channel Assignment Strategies, Handoff Strategies- Prioritizing Handoffs, Practical Handoff Considerations, Trunking and Grade of Service							
UNIT-II (10 Hrs.)	Mobile Radio Propagation: Large-Scale Path Loss: Introduction to Radio Wave Propagation, Free Space Propagation Model, Relating Power to Electric Field, Basic Propagation Mechanisms, Reflection: Reflection from Dielectrics, Brewster Angle, Reflection from perfect conductors, Ground Reflection (Two-Ray) Model, Diffraction: Fresnel Zone Geometry, Knife-edge Diffraction Model, Multiple knife-edge Diffraction, Scattering, Outdoor Propagation Models- Longley-Ryze Model, Okumura Model, Hata Model, PCS Extension to Hata Model, Walfisch and Bertoni Model, Wideband PCS Microcell Model, Indoor Propagation Models Partition losses (Same Floor), Partition losses between Floors, Log-distance path loss model, Ericsson Multiple Breakpoint Model, Attenuation Factor Model, Signal penetration into buildings, Ray Tracing and Site Specific Modeling.							
UNIT-III (8 Hrs.)	Mobile Radio Propagation: Small –Scale Fading and Multipath Small Scale Multipath propagation-Factors influencing small scale fading, Doppler shift, Impulse Response Model of a multipath channel- Relationship between Bandwidth and							

	Received power, Small Scale Multipath Measurements-Direct RF Pulse System, Spread Spectrum Sliding Correlator Channel Sounding, Frequency Domain Channels Sounding, Parameters of Mobile Multipath Channels-Time Dispersion Parameters, Coherence Bandwidth, Doppler Spread and Coherence Time, Types of Small Scale Fading-Fading effects Due to Multipath Time Delay Spread, Flat fading, Frequency selective fading, Fading effects Due to Doppler Spread-Fast fading, slow fading, Statistical Models for multipath Fading Channels-Clarke's model for flat fading, spectral shape due to Doppler spread in Clarke's model, Simulation of Clarke and Gans Fading Model, Level crossing and fading statistics, Two-ray Rayleigh Fading Model.
UNIT-IV (10 Hrs.)	Equalization and Diversity Introduction, Fundamentals of Equalization, Training a Generic Adaptive Equalizer, Equalizers in a communication Receiver, Linear Equalizers, Non-linear Equalization-Decision Feedback Equalization (DFE), Maximum Likelihood Sequence Estimation (MLSE) Equalizer, Algorithms for adaptive equalization-Zero Forcing Algorithm, Least Mean Square Algorithm, Recursive least squares algorithm. Diversity -Derivation of selection Diversity improvement, Derivation of Maximal Ratio Combining improvement, Practical Space Diversity Consideration-Selection Diversity, Feedback or Scanning Diversity, Maximal Ratio Combining, Equal Gain Combining, Polarization Diversity, Frequency Diversity, Time Diversity, RAKE Receiver.
UNIT-V (10 Hrs.)	Wireless Networks: Introduction to wireless Networks, Advantages and disadvantages of Wireless Local Area Networks, WLAN Topologies, WLAN Standard IEEE 802.11, IEEE 802.11 Medium Access Control, Comparison of IEEE 802.11 a,b,g and n standards, IEEE 802.16 and its enhancements, Wireless PANs, Hyper Lan, WLL.
Text Books:	
1.	Wireless Communications, Principles, Practice – Theodore, S. Rappaport, 2 nd Ed., 2002, PHI.
2.	Wireless Communications-Andrea Goldsmith, 2005 Cambridge University Press.
3.	Mobile Cellular Communication – Gottapu Sasibhushana Rao, Pearson Education, 2012
Reference Book:	
1.	Principles of Wireless Networks – KavehPahlavan and P. Krishna Murthy, 2002, PE
2.	Wireless Digital Communications – KamiloFeher, 1999, PHI.
3.	Wireless Communication and Networking – William Stallings, 2003, PHI.
4.	Wireless Communication – UpenDalal, Oxford Univ.
5.	Press Wireless Communications and Networking – Vijay K. Gary, Elsevier.

Code	Category	L	T	P	C	I.M	E.M	Exam
M19CS1202	PC	3	0	0	3	25	75	3 Hrs.
IMAGE AND VIDEO PROCESSING								
Course Objectives:								
1	The basic concepts and methods to develop foundation in digital image processing and video processing are introduced and The Importance of various image transforms, image transform properties are discussed.							
2	Understanding the image enhancement techniques in both spatial domain and frequency domain.							
3	The process of recovering image that has been degraded by noise or any other degradation phenomenon.							
4	Understanding the importance of image segmentation and various methods used for segmentation, The importance of reducing the data for digital image representation by using various image compression techniques							
5	To understand the importance of video processing in multimedia and the various video formation models, motion estimation techniques in video processing							
6	Applications of motion estimation in video processing							
Course Outcomes								
S.No	Outcome							Knowledge Level
1	Defining the digital image, representation of digital image, importance of image resolution, applications in image processing.							K2
2	Know the advantages of representation of digital images in transform domain, application of various image transforms							K2
3	Know how an image can be enhanced by using histogram techniques, filtering techniques etc.							K3
4	Understand image degradation, image restoration techniques using spatial filters and frequency domain							K3
5	Know the detection of point, line and edges in images, edge linking through local processing, global processing.							K3
6	Understand the redundancy in images, various image compression techniques.							K2
7	Know the video technology from Analog colour TV systems to digital video systems, how video signal is sampled and filtering operations in video processing.							K2
8	Know the general methodologies for 2D motion estimation, various coding used in video processing							K3
SYLLABUS								
UNIT-I (10 Hrs)	Fundamentals of Image Processing and Image Transforms: Introduction, Image sampling, Quantization, Resolution, Image file formats, Elements of image processing system, Applications of Digital image processing Introduction, Need for transform, image transforms, Fourier transform, 2 D Discrete Fourier transform and its transforms, Importance of phase, Walsh transform, Hadamard transform, Haar transform, slant transform Discrete cosine transform, KL transform, singular value decomposition, Radon transform, comparison of different image transforms.							
UNIT-II (10 Hrs)	Image Enhancement: Spatial domain methods: Histogram processing, Fundamentals of Spatial filtering, Smoothing spatial filters, Sharpening spatial filters. Frequency domain methods: Basics of filtering in frequency domain, image smoothing, image sharpening, Selective filtering.							

	Image Restoration: Introduction to Image restoration, Image degradation, Types of image blur, Classification of image restoration techniques, Image restoration model, Linear and Nonlinear image restoration techniques, Blind deconvolution.
UNIT-III (10 Hrs)	Image Segmentation: Introduction to image segmentation, Point, Line and Edge Detection, Region based segmentation., Classification of segmentation techniques, Region approach to image segmentation, clustering techniques, Image segmentation based on thresholding, Edge based segmentation, Edge detection and linking, Hough transform, Active contour Image Compression: Introduction, Need for image compression, Redundancy in images, Classification of redundancy in images, image compression scheme, Classification of image compression schemes, Fundamentals of information theory, Run length coding, Shannon – Fano coding, Huffman coding, Arithmetic coding, Predictive coding, Transformed based compression, Image compression standard, Wavelet-based image compression, JPEG Standards.
UNIT-IV (8 Hrs)	Basic Steps of Video Processing: Analog Video, Digital Video. Time-Varying Image Formation models: Three-Dimensional Motion Models, Geometric Image Formation, Photometric Image Formation, Sampling of Video signals, Filtering operations.
UNIT-V (10 Hrs)	2-D Motion Estimation: Optical flow, General Methodologies, Pixel Based Motion Estimation, Block- Matching Algorithm, Mesh based Motion Estimation, Global Motion Estimation, Region based Motion Estimation, Multi resolution motion estimation, Waveform based coding, Block based transform coding, Predictive coding, Application of motion estimation in Video coding.
Text Books:	
1.	Digital Image Processing – Gonzaleze and Woods, 3rd Ed., Pearson.
2.	Video Processing and Communication – Yao Wang, JoemOstermann and Ya–quin Zhang. 1st Ed., PH Int.
3.	S.Jayaraman, S.Esakkirajan and T.VeeraKumar, “Digital Image processing, Tata McGraw Hill publishers, 2009
Reference Books:	
1.	Digital Image Processing and Analysis-Human and Computer Vision Application with CVIP Tools – ScotteUmbaugh, 2nd Ed, CRC Press, 2011.
2.	Digital Video Processing – M. Tekalp, Prentice Hall International.
3.	Digital Image Processing – S.Jayaraman, S.Esakkirajan, T.Veera Kumar – TMH, 2009.
4.	Multidimentional Signal, Image and Video Processing and Coding – John Woods, 2nd Ed, Elsevier.
5.	Digital Image Processing with MATLAB and Labview – Vipula Singh, Elsevier.
6.	Video Demystified – A Hand Book for the Digital Engineer – Keith Jack, 5th Ed., Elsevier.

Code	Category	L	T	P	C	I.M	E.M	Exam
M19 CS 1203	PE	3	0	0	3	25	75	3 Hrs.
SOFT COMPUTING TECHNIQUES								
(Program Elective-III)								
Course Objectives:								
1	To study the soft computing basics							
2	To Understand the basic implementation details on Artificial Neural Networks							
3	To understand fuzzy logic and its application in ANN.							
4	To study the Genetic algorithm and its application							
5	To study the applications of Soft Computing.							
Course Outcomes								
S.No	Outcome							Knowledge Level
1	Understand the basic concepts of Artificial neural network systems.							K2
2	Understand the McCulloch-Pitts neuron model, simple and multilayer Perception, Adeline and Madeline concepts							K2
3	Data processing, Hopfield and self-organizing network.							K3
4	Difference between crisp sets to fuzzy sets, fuzzy models, fuzzification, inference, membership functions, rule based approaches and defuzzification.							K2
5	Self – organizing fuzzy logic control, nonlinear time delay systems.							K2
6	Understand the concept of Genetic Algorithm steps. Tabu, ant-colony search techniques for solving optimization problems.							K3
7	GA applications to power system optimization problems, identification and control of linear and nonlinear dynamic systems using MATLAB-Neural network toolbox.							K3
8	Know the application and importance of stability analysis							K3
SYLLABUS								
UNIT-I (10 Hrs.)	Introduction: Approaches to intelligent control, Architecture for intelligent control, Symbolic reasoning system, Rule-based systems, the AI approach, Knowledge representation - Expert systems.							
UNIT-II (10 Hrs.)	Artificial Neural Networks: Concept of Artificial Neural Networks and its basic mathematical model, McCulloch-Pitts neuron model, simple perceptron, Adaline and Madaline, Feed-forward Multilayer Perceptron, Learning and Training the neural network, Data Processing: Scaling, Fourier transformation, principal-component analysis and wavelet transformations, Hopfield network, Self-organizing network and Recurrent network, Neural Network based controller.							
UNIT-III (10 Hrs.)	Fuzzy Logic System: Introduction to crisp sets and fuzzy sets, basic fuzzy set operation and approximate reasoning, Introduction to fuzzy logic modelling and control, Fuzzification, inferencing and defuzzification, Fuzzy knowledge and rule bases, Fuzzy modelling and control schemes for nonlinear systems, Self-organizing fuzzy logic control, Fuzzy logic control for nonlinear time delay system.							
UNIT-IV (10 Hrs.)	Genetic Algorithm: Basic concept of Genetic algorithm and detail algorithmic steps, Adjustment of free parameters, Solution of typical control problems using genetic algorithm, Concept on some other search techniques like Tabu search and ant-colony search techniques for solving optimization problems.							

UNIT-V (8 Hrs.)	Applications: GA application to power system optimisation problem, Case studies: Identification and control of linear and nonlinear dynamic systems using MATLAB-Neural Network toolbox, Stability analysis of Neural Network interconnection systems, Implementation of fuzzy logic controller using MATLAB fuzzy-logic toolbox, Stability analysis of fuzzy control systems.
Text Books:	
1.	Introduction to Artificial Neural Systems - Jacek.M.Zurada, Jaico Publishing House, 1999.
2.	Neural Networks and Fuzzy Systems - Kosko, B., Prentice-Hall of India Pvt. Ltd., 1994.
Reference Books:	
1.	Fuzzy Sets, Uncertainty and Information - Klir G.J. & Folger T.A., Prentice-Hall of India Pvt. Ltd., 1993.
2.	Fuzzy Set Theory and Its Applications - Zimmerman H.J. Kluwer Academic Publishers, 1994.
3	Introduction to Fuzzy Control - Driankov, Hellendroon, Narosa Publishers.
4	Artificial Neural Networks - Dr. B. Yagananarayana, 1999, PHI, New Delhi.
5	Elements of Artificial Neural Networks - Kishan Mehrotra, Chelkuri K. Mohan, Sanjay Ranka, Penram International.
6	Artificial Neural Network –Simon Haykin, 2 nd Ed., Pearson Education.
7	Introduction Neural Networks Using MATLAB 6.0 - S.N. Shivanandam, S. Sumati, S. N. Deepa, 1/e, TMH, New Delhi.

Code	Category	L	T	P	C	I.M	E.M	Exam
M19 CS 1204	PE	3	0	0	3	25	75	3 Hrs.
INTERNET PROTOCOLS								
(Program Elective-III)								
Course Objectives:								
1	To provide theoretical and practical knowledge about the TCP/IP networking, which are fundamental to understand the architecture of the Internet.							
2	To explore Concepts like switching, multiplexing, naming, addressing, routing, quality of service, flow and congestion control through their implementation in TCP/IP protocol suite.							
Course Outcomes								
S.No	Outcome							Knowledge Level
1	Understanding basic network routing concepts and algorithms							K2
2	Understanding how to apply them into given topologies							K3
3	Understanding how the Internet protocol suite operates and describe the functions of various protocols							K1
4	Explain the concept and usage of node addressing and classify addresses into network layers.							K3
SYLLABUS								
UNIT-I (10 Hrs.)	Internetworking Concepts: Principles of Internetworking, Connectionless Internetworking, Application level Interconnections, Network level Interconnection, Properties of the Internet, Internet Architecture, Wired LANS, Wireless LANs, Point-to-Point WANs, Switched WANs, Connecting Devices, TCP/IP Protocol Suite. IP Address: Class full Addressing: Introduction, Class full Addressing, Other Issues, Sub-netting and Super-netting Classless Addressing: Variable length Blocks, Sub-netting, Address Allocation. Delivery, Forwarding, and Routing of IP Packets: Delivery, Forwarding, Routing, Structure of Router. ARP and RARP: ARP, ARP Package, RARP.							
UNIT-II (10 Hrs.)	Internet Protocol (IP): Datagram, Fragmentation, Options, Checksum, IP V.6. Transmission Control Protocol (TCP): TCP Services, TCP Features, Segment, A TCP Connection, State Transition Diagram, Flow Control, Error Control, Congestion Control, TCP Times. Stream Control Transmission Protocol (SCTP): SCTP Services, SCTP Features, Packet Format, Flow Control, Error Control, Congestion Control. Mobile IP: Addressing, Agents, Three Phases, Inefficiency in Mobile IP. Classical TCP Improvements: Indirect TCP, Snooping TCP, Mobile TCP, Fast Retransmit/ Fast Recovery, Transmission/ Time Out Freezing, Selective Retransmission, Transaction Oriented TCP.							
UNIT-III (10 Hrs.)	Unicast Routing Protocols (RIP, OSPF, and BGP): Intra and Inter-domain Routing, Distance Vector Routing, RIP, Link State Routing, OSPF,							

	Path Vector Routing, BGP. Multicasting and Multicast Routing Protocols: Unicast - Multicast- Broadcast, Multicast Applications, Multicast Routing, Multicast Link State Routing: MOSPF, Multicast Distance Vector: DVMRP.
UNIT-IV (10 Hrs.)	Domain Name System (DNS): Name Space, Domain Name Space, Distribution of Name Space, and DNS in the internet. Remote Login TELNET: Concept, Network Virtual Terminal (NVT). File Transfer FTP and TFTP: File Transfer Protocol (FTP). Electronic Mail: SMTP and POP. Network Management-SNMP: Concept, Management Components, World Wide Web-HTTP Architecture.
UNIT-V (8 Hrs.)	Multimedia: Digitizing Audio and Video, Network security, security in the internet firewalls. Audio and Video Compression, Streaming Stored Audio/Video, Streaming Live Audio/Video, Real-Time Interactive Audio/Video, RTP, RTCP, Voice Over IP. Network Security, Security in the Internet, Firewalls.
Text Books:	
1.	TCP/IP Protocol Suite- Behrouz A. Forouzan, Third Edition, TMH
2.	Internetworking with TCP/IP Comer 3 rd edition PHI
Reference Books:	
1.	High performance TCP/IP Networking- Mahbub Hassan, Raj Jain, PHI, 2005
2.	Data Communications & Networking – B.A. Forouzan– 2nd Edition – TMH
3.	High Speed Networks and Internets- William Stallings, Pearson Education, 2002
4.	Data and Computer Communications, William Stallings, 7th Edition., PEI.
5.	The Internet and Its Protocols – AdrinFarrel, Elsevier, 2005.

Code	Category	L	T	P	C	I.M	E.M	Exam
M19 CS 1205	PE	3	0	0	3	25	75	3 Hrs.
CYBER SECURITY								
(Program Elective-III)								
Course Objectives:								
1.	To gain knowledge about securing both clean and corrupted systems, protect personal data, and secure computer networks.							
2.	To understand principles of web security.							
3.	To gain familiarity with prevalent network and distributed system attacks, defences against them, and forensics to investigate the aftermath.							
4.	To gain an understanding of cryptography, how it has evolved, and some key encryption techniques used today.							
5.	To incorporate approaches for incident analysis and response.							
6.	To develop an understanding of security policies (such as confidentiality, integrity, and availability), as well as protocols to implement such policies.							
Course Outcomes								
S.No	Outcome							Knowledge Level
1	Understand Cyber Security architecture principles							K2
2	Identify System and application security threats and vulnerabilities							K1
3	Identify different classes of attacks							K1
4	Cyber Security incidents to apply appropriate response							K3
5	Describing risk management processes and practices							K1
6	Evaluation of decision making outcomes of Cyber Security scenario							K3
SYLLABUS								
UNIT-I (10 Hrs)	Introduction: Security Attacks (Interruption, Interception, Modification and Fabrication), Security Services (Confidentiality, Authentication, Integrity, Non-repudiation, access Control and Availability) and Mechanisms, A model for Internetwork security, Internet Standards and RFCs, Buffer overflow & format string vulnerabilities, TCP session hijacking, ARP attacks, route table modification, UDP hijacking, and man-in-the-middle attacks.							
UNIT-II (10 Hrs)	Conventional Encryption: Conventional Encryption Principles, Conventional encryption algorithms, cipher block modes of operation, location of encryption devices, key distribution Approaches of Message Authentication, Secure Hash Functions and HMAC							
UNIT-III (10 Hrs)	Number Theory: Prime and Relatively Prime Numbers, Modular Arithmetic, Fermat’s and Euler’s Theorems, The Chinese Remainder theorem, Discrete logarithms Public key: Public key cryptography principles, public key cryptography algorithms, digital signatures, digital Certificates, Certificate Authority and key management Kerberos, X.509 Directory Authentication Service							

UNIT-IV (8 Hrs)	IP Security: IP Security Overview, IP Security Architecture, Authentication Header, Encapsulating Security Payload, Combining Security Associations and Key Management Transport Level Security: Web Security Requirements, Secure Socket Layer (SSL) and Transport Layer Security (TLS), Secure Electronic Transaction (SET) Email Privacy: Pretty Good Privacy (PGP) and S/MIME.
UNIT-V (10 Hrs)	Intrusion Detection: Intruders, Intrusion Detection systems, Password Management. Malicious Software: Viruses and related threats & Countermeasures. Fire walls: Firewall Design principles, Trusted Systems.
Text Books:	
1.	Network Security & Cryptography: Principles and Practices, William Stallings, PEA, Sixth edition.
2.	Hack Proofing your Network, Russell, Kaminsky, Forest Puppy, Wiley Dreamtech
Reference Books:	
1.	Network Security & Cryptography, Bernard Menezes, Cengage, 2010

Code	Category	L	T	P	C	I.M	E.M	Exam
M19 CS 1206	PC	3	0	0	3	25	75	3 Hrs.
OPTICAL NETWORKS								
(Program Elective-IV)								
Course Objectives: To learn about								
1.	Various components of optical networks							
2.	Multiplexing techniques and fibre characteristics							
3.	First generation and broadcast optical network							
4.	Network management and access networks							
5.	Various photonic switches							
Course Outcomes								
S.No	Outcome							Knowledge Level
1	Able to Contribute in the areas of optical network and WDM network design.							K3
2	Able to Implement simple optical network and understand further technology developments for future enhanced network.							K3
SYLLABUS								
UNIT-I (10 Hrs.)	Client Layers of Optical Networks: SONET / SDH – Multiplexing, Frame Structure, Physical Layer, Infrastructure, ATM – Functions, Adaptation layers, QoS, Flow Control Signaling and Routing, IP – Routing, QoS, MPLS, Storage Area Networks – ESCON, Fiber Channel, HIPPI, Gigabit Ethernet.							
UNIT-II (10 Hrs.)	WDM network Elements and Design: Optical Line Terminals and Amplifiers, Add/Drop Multiplexers, Optical Cross Connects, Cost trade-offs in Network Design, LTD and RWA Problems, Dimensioning – Wavelength Routing Networks, Statistical and Maximum Load Dimensioning Models.							
UNIT-III (10 Hrs.)	Network Control and Management: Network Management Functions, Optical Layer Services and Interfacing, Layers within Optical Layer, Multivendor Interoperability, Performance and Fault Management, Configuration Management, Optical Safety.							
UNIT-IV (10 Hrs.)	Network Survivability: Basic Concepts of Survivability, Protection in SONET/SDH Links and Rings, Protection in IP Networks, Optical Layer Protection – Service Classes, Protection Schemes, Interworking between Layers.							
UNIT-V (8 Hrs.)	Access Networks and Photonic Packet Switching: Network Architecture, Enhanced HFC, FTTC, Photonic Packet Switching – OTDM, Synchronization, Header Processing, Buffering, Burst Switching, Test Beds.							
Text Books:								
1.	Optical Networks: A Practical Perspective - Rajiv Ramaswami and Kumar N. Sivarajan, 2 nd Ed., 2004, Elsevier Morgan Kaufmann Publishers (An Imprint of Elsevier).							
2.	WDM Optical Networks: Concepts, Design and Algorithms – C. Siva Rama Murthy and Mohan Guruswamy 2 nd Ed., 2003, PEI.							
3.	Optical Networks: Third Generation Transport Systems – Uyles Black, 2 nd Ed., 2009, PEI.							

Reference Books:	
1.	Optical Fiber Communications: Principles and Practice – John.M.Senior, 2 nd Ed., 2000, PE.
2.	Fiber Optics Communication – Harold Kolimbris, 2 nd Ed., 2004, PEI.
3.	Networks – Timothy S. Ramteke, 2 ed., 2004, PEI.
4.	Optical Fiber Communications – GovindAgarwal, 2 nd Ed., 2004, TMH.
5.	Optical Fiber Communications and Its Applications – S.C.Gupta, 2004, PHI.
6.	Telecommunication System Engineering –Roger L.Freeman, 4 th Ed., John Wiley, 2004.

Code	Category	L	T	P	C	I.M	E.M	Exam
M19 CS 1207	PE	3	0	0	3	25	75	3 Hrs.
DIGITAL SIGNAL PROCESSORS AND ARCHITECTURES								
(Program Elective-IV)								
Course Objectives:								
1.	To study the basic DFT, FFT and rate conversion algorithms.							
2.	To study the number format, dynamic range and sources of errors in DSP systems.							
3.	To learn about TMS programmable DSPs and their programming capabilities.							
4.	To understand basic DSP algorithms on TMS processor.							
5.	To study the FFT algorithms on TMS320C54XX DSP device.							
Course Outcomes								
S.No	Outcome							Knowledge Level
	At the end of this course, students will be able to							
1	Identify and formalize architectural level characterization of P-DSP hardware							K1
2	Ability to design, programming (assembly and C), and testing code using Code Composer Studio environment							K3
3	Deployment of DSP hardware for Control, Audio and Video Signal processing applications							K3
4	Understanding of major areas and challenges in DSP based embedded systems.							K1
SYLLABUS								
UNIT-I (10 Hrs)	Introduction to Digital Signal Processing: Introduction, A Digital signal-processing system, The sampling process, Discrete time sequences. Discrete Fourier Transform (DFT) and Fast Fourier Transform (FFT), Linear time-invariant systems, Digital filters, Decimation and interpolation. Computational Accuracy in DSP Implementations: Number formats for signals and coefficients in DSP systems, Dynamic Range and Precision, Sources of error in DSP implementations, A/D Conversion errors, DSP Computational errors, D/A Conversion Errors, Compensating filter.							
UNIT-II (10 Hrs)	Architectures for Programmable DSP Devices: Basic Architectural features, DSP Computational Building Blocks, Bus Architecture and Memory, Data Addressing Capabilities, Address Generation UNIT, Programmability and Program Execution, Speed Issues, Features for External interfacing.							
UNIT-III (10 Hrs)	Programmable Digital Signal Processors: Commercial Digital signal-processing Devices, Data Addressing modes of TMS320C54XX DSPs, Data Addressing modes of TMS320C54XX Processors, Memory space of TMS320C54XX Processors, Program Control, TMS320C54XX instructions and Programming, On-Chip Peripherals, Interrupts of TMS320C54XX processors, Pipeline operation of TMS320C54XX Processors.							
UNIT-IV (8 Hrs)	Analog Devices Family of DSP Devices: Analog Devices Family of DSP Devices – ALU and MAC block diagram, Shifter Instruction, Base Architecture of ADSP 2100, ADSP-2181 high performance Processor. Introduction to Black fin Processor - The Black fin Processor, Introduction to Micro Signal Architecture, Overview of Hardware Processing Units and Register files, Address							

	Arithmetic Unit, Control Unit, Bus Architecture and Memory, Basic Peripherals.
UNIT-V (10 Hrs)	Interfacing Memory and I/O Peripherals to Programmable DSP Devices: Memory space organization, External bus interfacing signals, Memory interface, Parallel I/O interface, Programmed I/O, Interrupts and I/O, Direct memory access (DMA).
Text Books:	
1.	Digital Signal Processing – Avtar Singh and S. Srinivasan, Thomson Publications, 2004.
2.	A Practical Approach to Digital Signal Processing - K Padmanabhan, R. Vijayarajeswaran, Ananthi. S, New Age International, 2006/2009
3.	EmbeddedSignalProcessingwiththeMicroSignalArchitecturePublisher: Woon-SengGan, Sen M. Kuo, Wiley-IEEE Press, 2007
Reference Books:	
1.	Digital Signal Processors, Architecture, Programming and Applications – B. Venkataramani and M. Bhaskar, 2002, TMH.
2.	Digital Signal Processing –Jonatham Stein, 2005, John Wiley.
3.	DSP Processor Fundamentals, Architectures & Features – Lapsley et al. 2000, S. Chand & Co.
4.	Digital Signal Processing Applications Using the ADSP-2100 Family by The Applications Engineering Staff of Analog Devices, DSP Division, Edited by Amy Mar, PHI
5.	<i>The Scientist and Engineer's Guide to Digital Signal Processing</i> by Steven W. Smith, Ph.D.,California Technical Publishing, ISBN 0-9660176-3-3, 1997
6.	Embedded Media Processing by David J. Katz and Rick Gentile of Analog Devices, Newnes , ISBN 0750679123, 2005

Code	Category	L	T	P	C	I.M	E.M	Exam
M19 CS 1208	PE	3	0	0	3	25	75	3 Hrs.
RADIO NAVIGATIONAL AIDS								
(Program Elective-IV)								
Course Objectives:								
1.	To learn about Navigational Systems and Inertial navigation							
2.	To acquire knowledge about and Global Positioning System and its applications in navigation							
3.	To become familiar with the concepts of Differential GPS and Wide Area Augmentation System							
Course Outcomes								
S.No	At the end of this course, students will be able to							Knowledge Level
1	Acquire knowledge about Navigational systems							K1
2	Analyse the concepts of GPS and its applications in Navigation							K3
3	Foster ability to work using Instrument Landing System.							K3
4.	Acquire knowledge about Satellite Navigation System.							K1
SYLLABUS								
UNIT-I (10 Hrs)	Navigational Systems: Review of Navigational Systems: Aircraft navigational system, Geometry of the earth. Navigation equation, Navigation errors, Radio navigation system types and Performance parameters, ILS System, Hyperbolic navigation systems, Loran, Omega, Decca Radio direction finding, DME, TACAN and VORTAC.							
UNIT-II (10 Hrs)	Inertial Navigation: Inertial navigation system ,Sensing instruments: Accelerometer. Gyro- copes, Analytic and Gimbaled platforms, Mechanization, Error analysis, Alignment.							
UNIT-III (10 Hrs)	Global Positioning System (GPS) for Navigation: Overview of GPS, Reference systems ,Satellite orbits, Signal structure, Geometric dilution of precision (GDOP), or Precision dilution of recision (PDOP), Satellite ephemeris, Satellite clock, Ionospheric group delay ,Tropospheric group delay, Multipath errors and Receiver measurement errors.							
UNIT-IV (10 Hrs)	Differential GPS and WAAS: Standard and precise positioning service local area DGPS and Wide area DGPS errors, Wide Area Augmentation System (WAAS) architecture, Link budget and Data Capacity, Ranging function, Precision approach and error estimates.							
UNIT-V (8 Hrs)	GPS Navigational Applications: General applications of GPS, DGPS, Marine, Air and Land Navigation, Surveying, Mapping and Geographical information systems, Military and Space							
Text Books:								
1.	Cryptography and Network Security: Principles and Practice - William Stallings, Pearson Education.							
2.	Parkinson. BW. Spilker - “Global Positioning System Theory and Applications”, Progress in Astronautics, Vol. I and II, 1996.							

Reference Books:	
1.	Hoffman. B., Wellenhof. H... Lichtenegger and J. Collins - "GPS Theory andPractice", Springer Verlag Wien New York, 1992.
2.	Elliot D. Kaplan - "Understanding GPS Principles and Applications", Artech House. Inc., 1996.
3.	Lieck Alfred. - "GPS Satellite Surveying", John Wiley, 1990.

Code	Category	L	T	P	C	I.M	E.M	Exam
M19 CS 1209	PC	0	0	4	2	25	75	3 Hrs.
ADVANCED COMMUNICATIONS LAB								
Course Objectives:								
1.	To apply the concepts of measurement of bit error rate, Hamming distance and coding techniques.							
2.	To simulate and calculate efficiency of spread spectrum techniques.							
3.	To implement filter response on DSP trainer kit.							
4.	To analyse the characteristics of Optical Fibre and Laser Diode.							
Course Outcomes								
S.No	At the end of this course, students will be able to							Knowledge Level
1.	Apply the concepts of measurement of bit error rate, Hamming distance and coding techniques.							K3
2.	Simulate and calculate efficiency of spread spectrum techniques.							K3
3.	Implement filter response on DSP trainer kit.							K3
4.	Analyse the characteristics of Optical Fibre and Laser Diode.							K4
Note:								
A. Minimum of 10 Experiments have to be conducted B. All Experiments may be Simulated using MATLAB and to be verified using related training kits								
List of Experiments								
1.	Measurement of Bit Error Rate using Binary Data							
2.	Verification of minimum distance in Hamming code							
3.	Determination of output of Convolutional Encoder for a given sequence							
4.	Determination of output of Convolutional Decoder for a given sequence							
5.	Efficiency of DS Spread- Spectrum Technique							
6.	Simulation of Frequency Hopping (FH) system							
7.	Effect of Sampling and Quantization of Digital Image							
8.	Verification of Various Transforms (FT / DCT/ Walsh / Hadamard) on a given Image (Finding Transform and Inverse Transform)							
9.	Point, Line and Edge detection techniques using derivative operators.							
10.	Implementation of FIR filter using DSP Trainer Kit (C-Code/ Assembly code)							
11.	Implementation of IIR filter using DSP Trainer Kit (C-Code/ Assembly code)							
12.	Determination of Losses in Optical Fibre							
13.	Observing the Waveforms at various test points of a mobile phone using Mobile Phone Trainer							
14.	Study of Direct Sequence Spread Spectrum Modulation & Demodulation using CDMA-DSS-BER Trainer							
15.	Study of ISDN Training System with Protocol Analyser							
16.	Characteristics of LASER Diode.							
Reference Books:								
1	Optical Fibre Communications: Principles and Practice – John.M.Senior, 2 nd Ed., 2000, PE.							
2	Data Communications and Networking, Behrouz A.Forouzan, 4 th Ed, Tata Mc-Graw Hill							
3	Digital communications, John G Proakis, Mc-Graw Hill							
4	Digital Signal Processing: Principles, Algorithms & Applications - J.G.Proakis, D. G. Manolakis, 4th Ed., PHI.							
5	DSP – A Practical Approach – Emmanuel C. Ifeachor, Barrie. W. Jervis, 2 Ed., Pearson Education							

Code	Category	L	T	P	C	I.M	E.M	Exam
M19 CS 1210	PC	0	0	4	2	25	75	3 Hrs.
ADVANCED DIGITAL IMAGE AND VIDEO PROCESSING LABORATORY								
Course Objectives:								
1.	To apply processing algorithms to enhance images and videos.							
2.	To apply compression and segmentation techniques of Images and Videos.							
3.	To apply object detection techniques in Image/ Video							
Course Outcomes								
S.No	Outcome							Knowledge Level
	At the end of this course, students will be able to							
1	Perform image and video enhancement							K3
2	Perform image and video segmentation							K3
3	Detect an object in an image/video							K3
LIST OF EXPERIMENTS								
1.	Perform basic operations on images like addition, subtraction etc.							
2.	Plot the histogram of an image and perform histogram equalization							
3.	Implement segmentation algorithms							
4.	Perform video enhancement							
5.	Perform video segmentation							
6.	Perform image compression using lossy technique							
7.	Perform image compression using lossless technique							
8.	Perform image restoration							
9.	Convert a colour model into another							
10.	Calculate boundary features of an image							
11.	Calculate regional features of an image							
12.	Detect an object in an image/video using template matching/Bayes classifier							
Text Books:								
1.	Digital Image Processing – Gonzaleze and Woods, 3rd Ed., Pearson.							
2.	Fundamentals of Digital Image Processing- Anil K. Jain, Prentice-Hall International Edition							
3.	S.Jayaraman, S.Esakkirajan and T.Veera Kumar, “Digital Image processing, Tata McGraw Hill publishers, 2009							
4.	Video Processing and Communication – Yao Wang, Joem Ostermann and Ya–quin Zhang. 1stEd., PH Int							
5.	Digital Video Processing – M. Tekalp, Prentice Hall International							
Reference Books:								
1.	Digital Image Processing with MATLAB and Lab view – Vipula Singh, Elsevier.							
2.	Multidimensional Signal, Image and Video Processing and Coding – John Woods, 2nd Ed, Elsevier.							
3.	Hand book of Image and video processing-AL Bovik, Academic Press							

Code	Category	L	T	P	C	I.M	E.M	Exam
M19 CS 1211	MC	0	0	4	2	100	--	3 Hrs.
MINI PROJECT WITH SEMINAR								
For Mini Project with Seminar, a student under the supervision of a faculty member, shall collect the literature on a topic and critically review the literature and submit it to the department in a report form and shall make an oral presentation before the Project Review Committee consisting of Head of the Department, supervisor/mentor and two other senior faculty members of the department. For Mini Project with Seminar, there will be only internal evaluation of 100 marks. A candidate has to secure a minimum of 50% of marks to be declared successful.								

Code	Category	L	T	P	C	I.M	E.M	Exam
#AC-2	AC	2	0	0	0	0	0	--
AUDIT COURSE-2								
List of Audit Courses and their Syllabi are mentioned in the First Semester Syllabus. The students can opt any one course for AC 2 from the list mentioned in first semester by not opting the course which is already taken for AC 1								



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(AUTONOMOUS)**
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Accredited by NAAC with 'A' Grade, All UG Programmes are accredited by NBA
CHINNA AMIRAM (P.O):: BHIMAVARAM :: W.G.Dt., A.P., INDIA :: PIN: 534 204

SCHEME OF INSTRUCTION & EXAMINATION
(Regulation R19)
M.TECH (COMMUNICATION SYSTEMS)
DEPARTMENT OF ELECTRONICS AND COMMUNICATION
ENGINEERING
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III-SEMESTER

Subject Code	Name of the Subject	Category	C	L	T	P	Internal Marks	External Marks	Total Marks
#PE-V	Program Elective	PE	3	3	--	--	25	75	100
#OE-I	Open Elective	OE	3	3	--	--	25	75	100
M19CS 2106	Dissertation-I /Industrial Project	PR	10	--	---	20	50	50	100
Total			16	6	0	20	100	200	300

	Course Code	Course
#PE-V/ MOOCS	M19CS2101	Detection and Estimation Theory
	M19CS2102	Coding Theory and Applications
	M19CS2103	Software Defined Radio
	M19 CS 2104 (MOOCS-I)	Students Going for Industrial Project / Thesis will complete these courses through MOOCS. Students can also choose SWAYAM or NPTEL with a 12 weeks' course duration in PG level with 3 credits, but the chosen subject should not be covered in their M. Tech Course
#OE-I/ MOOCS	#OE-I	Students have to choose one open elective course offered by departments other than the parent department. List of open Electives offered by other departments are enclosed.
	M19 CS 2105 (MOOCS-II)	Students Going for Industrial Project / Thesis will complete these courses through MOOCS. Students can also choose SWAYAM or NPTEL with a 12 weeks' course duration in PG level with 3 credits, but the chosen subject should not be covered in their M. Tech Course

OPEN ELECTIVES OFFERED TO OTHER DEPARTMENTS	
M19CS2107	Signals and systems
M19CS2108	Principles of Communication
M19CS2109	Image and video Processing

Code	Category	L	T	P	C	I.M	E.M	Exam
M19 CS 2101	PE	3	0	0	3	25	75	3 Hrs.
DETECTION AND ESTIMATION THEORY								
(Program Elective-V)								
Course Objectives:								
1.	To enable the students to acquire the fundamental concepts of Signal Detection and Estimation							
2.	To get familiarize with different Hypotheses in detection and estimation problems							
3.	To introduce the methods of Detection and estimation of signals in white and non-white Gaussian noise.							
4.	To familiarize with the detection of random signals.							
5.	To enable the students to understand the time varying waveform detection and its estimation							
Course Outcomes								
S.No	Outcome							Knowledge Level
1	Understand the basic concepts of signal detection and estimation							K2
2	Understand different hypotheses in detection and estimation problems							K2
3	Understand the conceptual basics of detection and estimation of signals in white and non-white Gaussian noise							K2
4	Understand the detection of random signals							K2
5	Understand the time varying waveform detection and its estimation							K2
SYLLABUS								
UNIT-I (10 Hrs)	Random Processes: Discrete Linear Models, Markov Sequences and Processes, Point Processes, and Gaussian Processes.							
UNIT-II (10 Hrs)	Detection Theory: Basic Detection Problem, Maximum A posteriori Decision Rule, Minimum Probability of Error Classifier, Bayes Decision Rule, Multiple-Class Problem (Bayes)- minimum probability error with and without equal a priori probabilities, Neyman-Pearson Classifier, General Calculation of Probability of Error, General Gaussian Problem, Composite Hypotheses							
UNIT-III (10 Hrs)	Linear Minimum Mean-Square Error Filtering: Linear Minimum Mean Squared Error Estimators, Nonlinear Minimum Mean Squared Error Estimators. Innovations, Digital Wiener Filters with Stored Data, Real-time Digital Wiener Filters, and Kalman Filters.							
UNIT-IV (8 Hrs)	Statistics: Measurements, Nonparametric Estimators of Probability Distribution and Density Functions, Point Estimators of Parameters, Measures of the Quality of Estimators, Introduction to Interval Estimates, Distribution of Estimators, Tests of Hypotheses, Simple Linear Regression, Multiple Linear Regression.							
UNIT-V (10 Hrs)	Estimating the Parameters of Random Processes from Data: Tests for Stationarity and Ergodicity, Model-free Estimation, Model-based Estimation of Autocorrelation Functions, Power Spectral Density Functions.							
Text Books:								
1.	Random Signals: Detection, Estimation and Data Analysis - K. Sam Shanmugan & A.M. Breipohl, Wiley India Pvt. Ltd, 2011.							

2.	Random Processes: Filtering, Estimation and Detection - Lonnie C. Ludeman, Wiley India Pvt. Ltd., 2010.
Reference Books:	
1.	Fundamentals of Statistical Signal Processing: Volume I Estimation Theory– Steven.M.Kay, Prentice Hall, USA, 1998.
2.	Fundamentals of Statistical Signal Processing: Volume I Detection Theory– Steven.M.Kay, Prentice Hall, USA, 1998.
3.	Introduction to Statistical Signal Processing with Applications - Srinath, Rajasekaran, Viswanathan, 2003, PHI.
4.	Statistical Signal Processing: Detection, Estimation and Time Series Analysis – Louis L.Scharf, 1991, Addison Wesley.
5.	Detection, Estimation and Modulation Theory: Part – I – Harry L. Van Trees, 2001, John Wiley & Sons, USA.
6.	Signal Processing: Discrete Spectral Analysis – Detection & Estimation – Mischa Schwartz, Leonard Shaw, 1975, McGraw Hill.

Code	Category	L	T	P	C	I.M	E.M	Exam
M19CS2102	PE	3	0	0	3	25	75	3 Hrs.
CODING THEORY AND APPLICATIONS								
(Program Elective-V)								
Course Objectives:								
1.	To study the concepts of information, entropy, mutual information and study the Shannon's fundamental limits, theorems on information transmission.							
2.	To Introduce and classify the Error correcting codes and understand the encoding and decoding of various linear block codes, Convolutional codes.							
3.	To Know the mathematical description of error correcting codes.							
4.	To Introduce the extension field called Galois field and their role in the design of BCH.							
5.	To Study the applications of error correcting codes.							
Course Outcomes								
S.No	Outcome							Knowledge Level
1	Analyze the information theoretic problems from various disciplines like computer science, mathematics, statistics and communication engineering.							K4
2	Apply coding techniques in various communication systems like wireless communications to achieve coding gain at low SNR values.							K6
3	Build new structures for encoder and decoder to address the issues in evaluating performance of communication system.							K6
4	Implement coding techniques in real time systems.							K6
5	Encode and decode information by applying algorithms associated with well-known codes.							K5
SYLLABUS								
UNIT-I (10 Hrs)								
Coding for Reliable Digital Transmission and Storage: Mathematical model of Information, A Logarithmic Measure of Information, Average Mutual Information and Entropy, Types of Errors, Error Control Strategies. Linear Block Codes: Introduction to Linear Block Codes, Syndrome and Error Detection, Minimum Distance of a Block code, Error-Detecting and Error-correcting Capabilities of a Block code, Standard array and Syndrome Decoding, Probability of an undetected error for Linear Codes over a BSC, Hamming Codes. Applications of Block codes for Error control in data storage system.								
UNIT-II (10 Hrs)								
Cyclic Codes: Description, Generator and Parity-check Matrices, Encoding, Syndrome Computation and Error Detection, Decoding, Cyclic Hamming Codes, Shortened cyclic codes.								
UNIT-III (10 Hrs)								
Convolutional Codes: Encoding of Convolutional Codes, Structural and Distance Properties of Convolutional codes, maximum likelihood decoding, Sequential decoding, Majority-logic decoding of Convolution codes, Applications of Viterbi Decoding and Sequential Decoding, Applications of Convolutional codes in ARQ system.								

UNIT-IV (8 Hrs)	Burst –Error-Correcting Codes: Decoding of Single-Burst Error Correcting Cyclic codes, Single – Burst – Error – Correcting Codes, Phased – Burst – Error – Correcting Codes, Burst – Error – Correcting Convolutional Codes, Bounds on Burst Error Correcting Capability, Interleaved Cyclic and Convolutional Codes.
UNIT-V (10 Hrs)	BCH- CODES Binary Primitive BCH Codes – Decoding of BCH Codes – Syndrome Computation and Iterative Algorithms for finding the Error Location Polynomial, Error location polynomial for single and double error correction, Computation of Error Location Numbers and Error Correction.
Text Books:	
1.	Error Control Coding- Fundamentals and Applications –Shu Lin, Daniel J.Costello, Jr, Prentice Hall, Inc.
2.	Digital Communications-Fundamentals and Application - Bernard Sklar, Pearson Education.
Reference Books:	
1.	Error Correcting Coding Theory-Man Young Rhee- 1989, McGraw-Hill Publishing
2.	Digital Communications- John G. Proakis, 5th Ed., 2008, TMH.
3.	Introduction to Error Control Codes-Salvatore Gravano-oxford
4.	Information Theory, Coding and Cryptography – Ranjan Bose, 2nd Ed, 2009, TMH.

Code	Category	L	T	P	C	I.M	E.M	Exam
M19CS 2103	PE	3	0	0	3	25	75	3 Hrs.
SOFTWARE DEFINED RADIO								
(Program Elective-V)								
Course Objectives:								
1.	To study the fundamental radio components and how these components are implemented in software.							
2.	To study the principles of software architecture to support and develop the SDR.							
3.	To study Policy and cooperation mechanisms that enable SDR to interoperate							
4.	To study and build SDR and investigate their role in future communication systems.							
Course Outcomes								
S.No	Outcome							Knowledge Level
1	Understanding of analog RF components as front end block in implementation of SDR.							K2
2	Design circuits at different Multirate signaling technique for frequency conversion and Sampling issues.							K6
3	Understanding of ADC and DAC technology.							K2
4	Knowledge of Hardware and software development methods for embedded wireless systems							K1
5	Make system-level decisions for software defined radio technology and products.							K5
SYLLABUS								
UNIT-I (10 Hrs)		Introduction: The Need for Software Radios, What is Software Radio, Characteristics and benefits of software radio- Design Principles of Software Radio, RF Implementation issues- The Purpose of RF Front – End, Dynamic Range- The Principal Challenge of Receiver Design – RF Receiver Front- End Topologies- Enhanced Flexibility of the RF Chain with Software Radios-Importance of the Components to Overall Performance- Transmitter Architectures and Their Issues- Noise and Distortion in the RF Chain, ADC and DAC Distortion.						
UNIT-II (10 Hrs)		Multi Rate Signal Processing: Introduction- Sample Rate Conversion Principles- Polyphase Filters- Digital Filter Banks- Timing Recovery in Digital Receivers Using Multirate Digital Filters. Digital Generation of Signals: Introduction- Comparison of Direct Digital Synthesis with Analog Signal Synthesis- Approaches to Direct Digital Synthesis- Analysis of Spurious Signals- Spurious Components due to Periodic jitter- Band Pass Signal Generation- Performance of Direct Digital Synthesis Systems- Hybrid DDS-PLL Systems- Applications of direct Digital Synthesis- Generation of Random Sequences- ROM Compression Techniques.						
UNIT-III (10 Hrs)		Analog to Digital and Digital to Analog Conversion: Parameters of ideal data converters- Parameters of Practical data converters- Analog to Digital and Digital to Analog Conversion- Techniques to improve data converter performance-Common ADC and DAC architectures.						

UNIT-IV (8 Hrs)	Digital Hardware Choices: Introduction- Key Hardware Elements- DSP Processors- Field Programmable Gate Arrays- Trade-Offs in Using DSPs, FPGAs, and ASICs- Power Management Issues- Using a Combination of DSPs, FPGAs, and ASICs.
UNIT-V (10 Hrs)	Object – Oriented Representation of Radios and Network Resources: Networks- Object Oriented Programming- Object Brokers- Mobile Application Environments-Joint Tactical Radio System. Case Studies in Software Radio Design: Introduction and Historical Perspective, SPEAKeasy- JTRS, Wireless Information Transfer System, SDR-3000 Digital Transceiver Subsystem, Spectrum Ware, CHARIOT.
Text Books:	
1.	Software Radio: A Modern Approach to Radio Engineering - Jeffrey H. Reed, 2002, PEA Publication.
2.	Software Defined Radio: Enabling Technologies- Walter Tuttle Bee, 2002, Wiley Publications.
Reference Books:	
1.	Software Defined Radio for 3G - Paul Burns, 2002, Artech House.
2	Software Defined Radio: Architectures, Systems and Functions - Markus Dillinger, KambizMadani, Nancy Alonistioti, 2003, Wiley.
3.	Software Radio Architecture: Object Oriented Approaches to wireless System Engineering – Joseph Mitola, III, 2000, John Wiley & Sons.
4.	R.F Microelectronics – B. Razavi, 1998, PHI.
5.	DSP – A Computer Based Approach – S. K. Mithra, 1998, McGraw-Hill

Code	Category	L	T	P	C	I.M	E.M	Exam
M19 CS 2104	PE	3	0	--	3	25	75	3 Hrs.
MOOCS-I								
Students Going for Industrial Project / Thesis will complete these courses through MOOCS. Students can also choose SWAYAM or NPTEL with a 12 weeks' course duration in PG level with 3 credits, but the chosen subject should not be covered in their M. Tech Course.								

Code	Category	L	T	P	C	I.M	E.M	Exam
	OE	--	--	--	3	25	75	3 Hrs.
OPEN ELECTIVE								
Students have to choose one open elective course offered by departments other than the parent departments. List of open Electives offered by other departments are given below.								

Offered from	Course Code	Course Name	Offered to
CIVIL ENGINEERING	M19 ST 2107	Construction Management	CST, CS, PSA,IT & CAD/CAM
	M19 ST 2108	Green Technology	
	M19 ST 2109	Analysis of Offshore Structures	
COMPUTER SCIENCE & ENGINEERING	M19 CST 2106	Python Programming	ST, CS, PSA & CAD/CAM
	M19 CST 2107	Artificial Intelligence	
	M19 CST 2108	Advanced Data structures	
ELECTRICAL & ELECTRONICS ENGINEERING	M19PS2107	Electric And Hybrid Vehicles	ST, CST, CS, IT & CAD/CAM
	M19PS2108	Energy From Waste	
	M19PS2109	Energy Management and Auditing	
INFORMATION TECHNOLOGY	M19IT2108	Web Technologies	ST, CS, PSA & CAD/CAM
	M19IT2109	Internet of Things	
	M19IT2110	Machine Learning	
MECHNAICAL ENGINEERING	M19CAD 2107	Operations Research	ST, CST, CS, PSA & IT
	M19CAD 2108	Nano Technology	
	M19CAD 2109	Product Design & Manufacturing	
SCIENCE & HUMANITIES	M19BS2101	Management and Organisational Behaviour	ST, CST, CS, PSA, IT & CAD/CAM

Code	Category	L	T	P	C	I.M	E.M	Exam
M19CS2105	OE	--	--	--	3	25	75	3 Hrs.
MOOCS-II								
Students Going for Industrial Project / Thesis will complete these courses through MOOCS. Students can also choose SWAYAM or NPTEL with a 12 weeks' course duration in PG level with 3 credits, but the chosen subject should not be covered in their M. Tech Course								

Code	Category	L	T	P	C	I.M	E.M	Exam
M19CS2106	PR	0	0	20	10	50	50	3 Hrs.
DISSERTATION-I/INDUSTRIAL PROJECT								
The Student has to register for Dissertation-I / Industrial project in III semester. Student has to submit, in consultation with his project supervisor, the title, objective and plan of action of his project work for approval. The student can initiate the Project work, only after obtaining the approval from the Project Review Committee (PRC). Continuous assessment of Dissertation-I during the III-Semester will be monitored by the PRC. Dissertation-I/ Industrial Project is evaluated for 50 internal marks and 50 external marks. Internal marks 50 awarded by Project Guide and PRC jointly based on continuous assessment consisting of two seminars based on Dissertation work-I. External marks 50 awarded by External Examiner, Supervisor and Head of the Department jointly based on a review and Viva voce on Dissertation work-I.								

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IV-SEMESTER

Subject Code.	Name of the Subject	Category.	C	L	T	P	Internal Marks	External Marks
M19CS2201	Dissertation-II / Industrial Project	PR	16	0	0	32	--	100

Code	Category	L	T	P	C	I.M	E.M	Exam
M19CS2201	PR	0	0	32	16	--	100	3 Hrs.
DISSERTATION-II/INDUSTRIAL PROJECT								
The student has to continue his/her work from Dissertation-I / Industrial project to complete Dissertation-II in IV semester. Continuous assessment of Dissertation-II during IV-Semester will be monitored by the PRC. Dissertation-II is evaluated for 100 external marks based on Review and Viva Voce. Review and Viva-Voce examination shall be conducted by a board consisting of the Supervisor, Head of the Department and the examiner who adjudicated the Thesis. The Board shall jointly report the candidate's work for 100 marks. If the report of the Viva-Voce is unsatisfactory (ie, < 50 marks), the candidate shall retake the Viva-Voce examination only after three months. If he fails to get a satisfactory report at the second Viva-Voce examination, the candidate has to reregister for the project and complete the project within the stipulated time after taking the approval from the College.								